

Higher Nationals – Summative Assignment Feedback Form

Student Name/ID	Nethmi Thiloka Divyanjalee (E170909)		
Unit Title	Unit: 6 Planning a Computing Project		
Assignment Number	1	Assessor	Mrs. Shenali
Submission Date	15 th of January 2025	Date Received 1st submission	24 th of October 2024
Re-submission Date		Date Received 2nd submission	

Assessor Feedback:

LO1. Conduct small-scale research, information gathering and data collection to generate knowledge on an identified subject

Pass, Merit & Distinction Descriptors P1 ☐ P2 ☐ M1 ☐ D1 ☐

LO2. Explore the features and business requirements of organisations in an identified sector

Pass, Merit & Distinction Descriptors P3 ☐ P4 ☐ M2 ☐

LO3. Produce project plans based on research of the chosen theme for an identified organisation

Pass, Merit & Distinction Descriptors P5 ☐ M3 ☐

LO4. Present your project recommendations and justifications of decisions made, based on research of the identified theme and sector

Pass, Merit & Distinction Descriptors P6 ☐ P7 ☐ P8 ☐ M4 ☐ D2 ☐

* Please note that grade decisions are provisional. They are only confirmed once internal and external moderation has taken place and grades decisions have been agreed at the assessment board.

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E170909
Planning a Computer Project - Assignment 01

Assessor Feedback:		
Grade:	Assessor Signature:	Date:
Resubmission Feedback: Please note resubmission feedback is focussed only on the resubmitted work		
Grade:	Assessor Signature:	Date:
Internal Verifier's Comments:		
Signature & Date:		

- Please note that grade decisions are provisional. They are only confirmed once internal and external moderation has taken place and grades decisions have been agreed at the assessment board.

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2. Avoid using page borders in your assignment body.
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4. Ensure that you give yourself enough time to complete the assignment by the due date.
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6. You must take responsibility for managing your own time effectively.
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11. If you are proven to be guilty of plagiarism or any academic misconduct, your grade could be reduced to A REFERRAL or at worst you could be expelled from the course
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Student name: Nethmi Thiloka Divyanjalee		Assessor name: Mrs. Shenali	
Issue date: 2024/10/24	Submission date: 2025/01/15	Submitted on: 2025/01/15	
Programme: BTEC HND in Computing			
Unit: 6 Planning a Computing Project			
Assignment number and title: 1 Managing a research project on practical applications and limitations of the use of AI			

Pearson Higher Nationals in Computing

Unit 6: Planning a Computing Project (PCP)

Assignment 01

Plagiarism

Plagiarism is a particular form of cheating. Plagiarism must be avoided at all costs and students who break the rules, however innocently, may be penalised. It is your responsibility to ensure that you understand correct referencing practices. As a university level student, you are expected to use appropriate references throughout and keep carefully detailed notes of all your sources of materials for material you have used in your work, including any material downloaded from the Internet. Please consult the relevant unit lecturer or your course tutor if you need any further advice.

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Student Declaration

Student declaration

I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.

Student signature e170909@esoft.academy  Date: 2025/01/15

Assignment Brief

Student Name/ID Number	Nethmi Thiloka Divyanjalee (E170909)
Unit Number and Title	Unit: 6 Planning a Computing Project
Academic Year	2024/2025
Unit Tutor	Mrs. Shenali
Assignment Title	Managing a research project on practical applications and limitations of the use of AI
Issue Date	2024/10/24
Submission Date	2025/01/15

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Planning a Computer Project - Assignment 01

Submission Format

This assignment should be submitted to the ELMS as a single document on or before the given submission date and time.

The document should comprise of

- **A comprehensive project plan** - including a work, time and resource allocation/ breakdown using appropriate tools. A business area analysis Including the features and operational areas of the business and the role of stakeholders and their impact on the success of the business.
- **A research paper** - including application and evaluation of quantitative and qualitative research methods to generate relevant primary data and examination of secondary sources to collect relevant secondary data and information.
- **An Action plan** – including recommendations and evaluation of project outcomes comparing the decisions given in the project plan.

The assignment should be word-processed, and figures can be either hand drawn or generated from a tool. There is no specific word limit for this assignment.

Unit Learning Outcomes

LO1 Conduct small-scale research, information gathering and data collection to generate knowledge on an identified subject

LO2 Explore the features and business requirements of organisations in an identified sector.

LO3 Produce project plans based on research of the chosen theme for an identified organisation

LO4 Present your project recommendations and justifications of decisions made, based on research of the identified theme and sector

Transferable skills and competencies developed

This unit offer students an opportunity to demonstrate the skills required for managing and implementing a project.

The students will undertake independent research and investigation for carrying out and executing a computing project which meets appropriate business aims and objectives.

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Students will develop competencies of:

- project management skills
- research skills
- writing, investigative, field study, performance skills, as appropriate presentation skills.

Assignment activity and guidance :

Research Theme: Artificial Intelligence

Research Topic: Practical applications and limitations of the use of AI in an identified sector (e.g. e-commerce, healthcare, biomedical science, education, lifestyle, automotive engineering, agriculture, gaming, finance)

“Artificial intelligence is at the forefront of innovation within Computer Science that uses a combination of logic, algorithms and large data sets to produce an AI model. The AI model is created to perform specific tasks or make predictions on supplied sets of input data, for example identifying patterns in weather data, internet search data or analysis of medical data. Artificial intelligence is predicted to generate a potential impact to the global economy of \$13 – \$15 Trillion by 2030, with sales of AI related hardware, software and services predicted to see a global revenue of \$900 billion. It is predicted that AI will boost the GDP of China by a little over 26% by 2039, and of North America by 14.5%. AI requires the input of structured and labelled data, where the outputs are already known. The input data sets to the AI model are intrinsically linked to study field to which the AI engine is to be applied. The AI model can then be used to identify and recognise patterns and relationships within the input data. This identification step is referred to as ‘training’ the AI model. Once this training is completed, the model can then be used to make predictions and identify patterns within brand new data sets. This new data set can then be added to the existing data set, so that the AI model keeps ‘growing’. As the model data set keeps expanding, and the AI algorithms are modified and refined, this gives the impression that the AI is ‘learning’ and demonstrating ‘intelligence’. AI has been used extensively to analyse and process large and complex datasets produced by big data systems, often in real time and using Computer Vision to extract data from image sources. Developing Artificial Intelligence required a range of knowledge and skills across a broad range of computer science disciplines. AI developers need to be familiar with the algorithms and techniques in fields such as machine learning, natural language processing, computer vision and data science. Knowing the required computing skills will help organisations recruit the correct resources to help develop and extend AI systems. Artificial Intelligence has a range of benefits across many industry sectors. In the finance industry, AI is rapidly becoming a game changer, using advanced algorithms, models and machine learning to carry out predictive analytics on large, rapidly

changing financial datasets to provide more accurate financial predictions. In the field of business operations, AI automation is helping to support and enhance labour productivity, leading to greater cost savings and increased efficiency. AI is also revolutionising the way businesses interact with their customers, by providing AI driven expert systems to help customers resolve queries as well as providing personalised recommendations based on customer choices and preferences. In the field of biomedical science, AI models help in the development of new drug treatments for a range of diseases by searching and processing large scale medical and DNA datasets. While Artificial Intelligence has numerous benefits in the analysis and processing of large data sets to solve problems, there are some clear risks to the application technology. AI systems respond to the data fed into the model, and so if this data is not representative of the problem area under study, there is a likelihood that the output of the AI model will be biased. In addition, there are security and privacy concerns on the source and storage of the large datasets used for AI. The rise of the Deepfake image and the manipulation of the human voice is also a concern because of the spread of misinformation. The wide-ranging effects of these risks mean that they can only be dealt with by a diverse range of stakeholders, including computer scientists, law makers, governments, and industry leaders. There are also incidental risks of AI in business, for example the increasing adoption of AI based systems may increase unemployment across a range of sectors and workforce demographics. Learners will have the opportunity to investigate the applications, benefits and limitations of Artificial Intelligence while exploring the responsibilities and solutions to the problems it is being used to solve.”. (Pearson, 2024)

You are expected to carry out small-scale research in order to explore the practical applications and limitations of the use of AI in an identified sector (e.g. e-commerce, healthcare, biomedical science, education, lifestyle, automotive engineering, agriculture, gaming, finance). The research that you carry out can be based on an organization that you have access to gather sufficient information to investigate the practical applications and limitations of the use of AI.

The findings of the research should be presented in a professionally compiled report and the report should cover the given tasks including,

- A comprehensive project plan - including a work, time and resource allocation/ breakdown using appropriate tools. A business area analysis Including the features and operational areas of the business and the role of stakeholders and their impact on the success of the business.

- A research paper - including application and evaluation of quantitative and qualitative research methods to generate relevant primary data and examination of secondary sources to collect relevant secondary data and information.
- An Action plan – including recommendations and evaluation of project outcomes comparing the decisions given in the project plan.

Activity 1 - A comprehensive project plan

Activity 1

1.1. Select an organization of your choice that allows you to explore and study the relevant data.

Provide an introduction and background to your project and the chosen organization. Define the scope and devise aims /objectives of the project that you are going to carry out. You also should include risks and benefits of exploring the practical applications and limitations of the use of AI of the chosen organization/s or the field.

2.1. Plan a small-scale research project to explore practical applications and limitations of the use of AI.

Produce a comprehensive project plan including the following.

- Cost, scope, time, quality, communication, risk, and resources management plan.
- Comprehensive Work Breakdown Structure (WBS) with clearly defined activities and milestones.
- Gantt chart to illustrate project schedule with realistic time allocated for each activity and clearly defined deadlines for milestones.

Activity 02 – Research Report

2.1. Discuss the features and operational areas of the chosen organization/s, the role and the impact of stakeholders for the success of the business. You also need analyse the challenges the organization/s may face in achieving the success and meeting business objectives by applying AI technologies to achieve operational efficiency.

2.2. Carry out the research to investigate the “**Practical applications and limitations of the use of AI in an identified sector** “ and generate relevant primary data by applying appropriate qualitative and quantitative research methods. You need to examine secondary sources to collect relevant secondary data and information to support the research. You then need to analyse the data and information and interpret the findings to generate knowledge on how the application of AI supports business requirements in the identified organization/s .

Activity 03: Action Plan

- 4.1 Communicate appropriate project recommendations derived from the research data analyzed for technical and non-technical audiences and assess the extent to which the project recommendations meet the needs of the chosen organization/s.
- 4.2 Discuss the reliability, accuracy, and the appropriateness of the research methods applied while arguing and evaluating the planning recommendations made in the project plan comparing them to the actual outcomes and the need of the chosen organization/s.

Recommended Resources

Please note that the resources listed are examples for you to use as a starting point in your research – the list is not definitive.

- Costley, C., Elliot, G. and Gibbs, P. (2010) Doing Work Based Research: Approaches to Enquiry for Insider-researchers. London: SAGE.
- Dawson, C. (2016) Projects in Computing and Information Systems: A Student's Guide. UK: Pearson Education.
- Gray, D. (2009) Doing Research in the Real World. 2nd Ed. London: SAGE.
- Guay, M., Schreiber, D. and Briones, S. (2016) The Ultimate Guide to Project Management: Learn everything you need to successfully manage projects and get them done. Free Kindle Edition. US: Zapier Inc.
- Lock, D. (2013) Project Management 8th Edition. UK: Routledge.
- Pinto, J.K. (2015) Project Management: Achieving Competitive Advantage 4th Ed. Pearson.

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Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Conduct small-scale research, information gathering and data collection to generate knowledge on an identified subject		D1 Interpret findings to generate knowledge on how the research theme supports business requirements in the identified sector.
P1 Demonstrate qualitative and quantitative research methods to generate relevant primary data for an identified theme. P2 Examine secondary sources to collect relevant secondary data and information for an identified theme.	M1 Analyse data and information from primary and secondary sources to generate knowledge on an identified theme.	
LO2 Explore the features and business requirements of organisations in an identified sector		
P3 Discuss the features and operational areas of a businesses in an identified sector. P4 Discuss the role of stakeholders and their impact on the success of a business.	M2 Analyse the challenges to the success of a business in an identified sector.	

Pass	Merit	Distinction
LO3 Produce project plans based on research of the chosen theme for an identified organisation		D2 Evaluate the project planning recommendations made in relation to the needs of the identified organisation and the accuracy and reliability of the research carried out.
P5 Devise comprehensive project plans for a chosen scenario, including a work and resource allocation breakdown using appropriate tools.	M3 Produce comprehensive project plans that effectively consider aims, objectives and risks/benefits for an identified organization.	
LO4 Present your project recommendations and justifications of decisions made, based on research of the identified theme and sector		
P6 Communicate appropriate project recommendations for technical and nontechnical audiences. P7 Present arguments for the planning decisions made when developing the project plans. P8 Discuss accuracy and reliability of the different research methods applied.	M4 Assess the extent to which the project recommendations meet the needs of the identified organisation, including fully supported rationales for planning decisions made.	

***** **END** *****

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Planning a Computer Project - Assignment 01

I'm Nethmi Thiloka Divyanjalee, and I'm a part of Kandy HND batch 103 at ESOFT Metro Campus. This is my 2nd semester. First, I would like to express our PCP Lecturer Mrs. Shenali who helped us to complete this assignment successfully. Thanks to this research, I was able to learn more about practical applications and limitations of the use of AI in an identified sector.

I would like to thank the ESOFT management for management for providing us with the necessary support that will facilitate us to learn and advance in knowledge.

Finally, I would like to thank my family and my dear friends for helping me in my work. They helped me to complete this assignment on time.

Thank You,
Nethmi Thiloka Divyanjalee,
E170909,
ESOFT Metro Campus,
Kandy.

The paper discusses the application of artificial intelligence in Araliya Bank online transaction platforms and how it would affect customer experience and satisfaction. The proposed study describes how Araliya Bank factors in this new area of AI-powered technologies to reimagine banking. Offering personalized financial solutions with transaction security and operational efficiency. The platform uses advanced algorithms in the delivery of real-time insights into financial management, automates customer support, and detects fraudulent activities to further drive seamless and intuitive banking.

The research highlights the platform's numerous benefits, such as improved transaction speed, tailored banking recommendations, round the clock service availability, and robust fraud prevention mechanisms. These features collectively contribute to great customer engagement, trust, and satisfaction. Additionally, the study explores the limitations of AI integration, including concerns over data privacy, system biases in decision making, and dependency on reliable technological infrastructure.

This paper, therefore, examines the extent to which Araliya Bank's AI platform caters to the needs of its customers while comparing some challenges in implementation and operation through qualitative and quantitative research methods. The findings underpin the transformative potential of AI within the banking sector, facilitating innovative solutions that are customer centric. However, the study instills that proper ethical data management, transparency, and enhancement of technologies for reduction of limitations are the points to be measured for better satisfaction of customers in the long term. This study contributes to the understanding of the evolving role of AI in digital banking and provides recommendations on how to maximize its benefits while minimizing associated risks.

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Activity 1 - A Comprehensive Project Plan

1. Introduction

1.1 Project Topic -: Practical applications and limitations of the use of AI in an Araliya Bank

1.2 Introduction to the Araliya Bank

As one of the premier financial services providers in Sri Lanka, Araliya Bank has always taken to embracing digital transaction to provide innovative and customer-centric banking solutions. It has developed an online transaction platform powered by AI, wherein state of the art artificial intelligence forms an intrinsic part for service enhancement seamlessly, efficiently, and in a secure way. This platform constitutes the backbone of the project in reviewing its role of improving customer experience and satisfaction by offering personalized financial solutions, automating transactions, and real-time assistance. With this in mind, Araliya Bank hopes to reduce complexity and time consumption traditionally associated with banking through the simplification of tasks and hence assure customers that modern banking is convenient, secure, and accessible anywhere and anytime.

The project undertaken critically analyzes the effectiveness of the AI powered platform in improving customer experience through the evaluation of various integrations of AI technologies, such as personalized recommendations, automated support, and faster transaction processing. The evaluation will also look at how the platform addresses such challenges as reducing transaction time, enhancing security measures, and providing customized financial advice. This project will find out the technical challenges the customer is facing to adapt, and also address some potential risks pertaining to data privacy. In this way, the leadership at Araliya Bank will have concrete, actionable insights for refinement of this platform and its features in the future toward adaptability in customer expectations.

The purpose of this project, therefore, is to discover how artificial intelligence will transform the face of banking with regard to customer service and the operations of banking itself. The project will be able to delve deeper into how AI will help Araliya Bank in meeting customer evolving needs, increasing customer engagement, and enhancing satisfaction levels. This research will also explore both the strengths and limitations of the platform, hence providing useful insights that can be further optimized. The

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recommendations presented will allow Araliya Bank to fine tune its digital strategy and stay competitive in an increasingly digital banking world, while acting as a yardstick for other financial institutions keen on embedding AI into their service delivery.

1.3 Overview of the Project Life Cycle and the Importance of Effective Project Management

Project management basically deals with the art of leading a project through predefined and structured phases from initiation towards completion. A series of phases through which a project usually passes includes its initiation to its closure. The life cycle of any project helps in ensuring that project objectives are accomplished effectively in a timely manner and within authorized cost. The project life cycle and importance of effective project management will be discussed here, considering key project management knowledge areas in relation to a project such as Araliya Bank's AI powered online transaction platform with a focus on enhancing customer experience through advanced technology.

1.3.1 Phases of the Project Life Cycle

The project life cycle usually consists of four major phases,

- Initiation
- Planning
- Execution
- Closure

Each phase has its tasks, deliverables, and results that add up to the overall success of the project.

- **Phase 1 -: Initiation**

The inception is the very beginning of the project, where the actual idea is created into a tangible plan. This mainly deals with the definition of objectives, scope, and purpose, besides identification of stakeholders. In a project like Araliya Bank's AI powered platform, this would involve validation of the business need for the AI system, identification of key stakeholders.

e.g., Customers, IT teams, bank management and alignment of project goals with the bank's overall business strategy.

This normally involves early key decisions about the viability of the project in terms of cost, time, and scope. Also at this stage, the development of a project charter or project brief may be developed which would describe the purpose of the project, goals, stakeholders, and deliverables. The outcome of this phase is approval to proceed with the project.

- **Phase 2 -: Planning**

Once the project is approved, the planning phase takes over. The success of the entire project is critical in the sense that it lays a platform for implementation. During this phase, the project manager and his team develop very detailed plans on how to execute the tasks involved in the project. The development of a project management plan, setting of deliverables, timelines, resource allocation, budgeting, risk management strategies, and communication plans are the various things involved in this phase.

In the strategy for Araliya Bank's AI power platform, this planning phase defines the scope of the work in great detail: identifying the exact technical requirements of the AI powered platform, setting milestones for infrastructure, and integrating the AI features. Also, a potential risk management plan will be developed to mitigate every potential risk arising, such as issues related to data privacy or some kind of resistance due to structural barriers to adopting AI.

- **Phase 3 -: Execution**

The execution phase is the particular stage of the project where the actual work is performed. This is where tasks are executed, resources are allocated, and the working team puts effort into the pre-defined objectives. This also covers monitoring and controlling for any deviations in the progress of the project.

During the execution phase, the development and implementation of Araliya Bank's AI platform will be done using AI technology such as machine learning algorithms and automated transaction systems. Implementation progress should be continuously monitored to keep the project on track regarding progress, utilization of the budget and use of

resources. Updates and reviews are regularly carried out with stakeholders to ensure that the projects remain within their goals and expectations.

- **Phase 4 -: Closure**

The closure phase means that the completion of the project. During the closing phase of a project, success of project, deliverables are handed over to the stakeholders and formally disbands the project team. A final project report would be produced, which document lessons learned, performance against original goals are prepared along with feedback from stakeholders.

Closure for Araliya Bank involves the final testing and rollout of the system to ensure all components of the AI powered platform are working as expected. The project team will go through a post implementation review to assess the impact on customer satisfaction, efficiency, and security brought in by the platform. It may also involve feedback from the customer and adjusting them for future improvements, if needed.

1.3.2 Importance of Effective Project Management

Effective Project Management is the lifeblood of any successful project, especially in complex and high stakes environments such as banking. Application of best practices and knowledge areas of project management will result in a much better chance that the project objectives will be met on time and within budget.

Important knowledge areas for project management

Several knowledge domains are necessary for effective project management in order to guarantee that the project is properly run and in line with its goals, some of the key areas of knowledge are listed below,

- **Integration Management**

Integration management ensures that all the aspects of the project are in good coordination. e.g., AI platform integration into the existing systems at Araliya Bank and alignment of the project team's work with the bank's strategic objectives.

- **Scope Management**

The scope management is a very critical initiative that will ensure the project delivers all the required deliverables with no scope creep. For Araliya Bank, this means setting boundaries on the AI platform's features that involve such capabilities as personalized recommendations, secure transactions, and automation of customer support, and ensuring these features are delivered within the defined scope.

- **Time Management**

Time management ensures the project is done in time. Setting realistic timelines for each phase and milestones in developing Araliya Bank's AI platform would avert any delay in the development that could impact customer satisfaction and overall success.

- **Cost Management**

Effective cost management ensures that the project remains within budget. While controlling costs to ensure the project does not go out of budget, it is necessary to deliver value to customers, the resource bank has to carefully allocate resources for AI technology, system integration, and training.

- **Quality Management**

It ensures that the required standards are met by the deliverables. Precisely, this will ensure for Araliya Bank's AI platform that the AI systems work correctly, securely, and reliably to offer a superior quality user experience that engenders customer trust.

- **Human Resource Management**

Human Resource Management should be focused on team building and leadership. The AI project at Araliya Bank can be successfully implemented only by bringing together the talented professionals' skills in AI development, IT, and customer service.

- **Risk Management**

It involves identification, analysis, and mitigation of risks. Some of the risks that may arise with Araliya Bank's AI platform are technological barriers, data privacy concerns, or customer adoption challenges. Effective risk management will help in identifying potential risks at an early stage and developing strategies to mitigate them.

- **Communication Management**

Communication management will ensure that stakeholders are kept duly informed. In the case of Araliya Bank, this means transparent communication among the internal teams and with customers and external vendors for seamless delivery of the AI platform.

- **Procurement Management**

Procurement Management will address how to attain resources and services. For the development of such a platform, Araliya Bank should buy certain AI technologies, software, and other services related products from vendors, therefore, good procurement management will become crucial.

- **Stakeholders Management**

Stakeholder management will ensure that the expectations of every stakeholder are met, from customers and employees to bank management. Stakeholders' satisfaction is very important in such a project like Araliya Bank's AI platform, which actually is targeted at enhancing customer experience.

The project life cycle provides a structured framework that guides a project from initiation through to closure. Each phase plays a vital role in ensuring the success of the project. Effective project management, supported by the key knowledge areas, ensures that the project is completed on time, within budget, and meets its objectives. In the online transaction platform using AI for Araliya Bank, best practice in project management will help deliver a solution that would offer an enhanced customer experience, better service efficiency, and facilitate the bank's digital transformation.

1.4 Project Objectives

The present research project is designed to review and assess the impact of Araliya Bank's new AI based online transaction platform on its customers to understand the strengths, limitations, and areas for improvement of this system. It is also designed to provide, in depth insight, optimize an online transaction platform for customers so that the objective of the bank would be achieved to provide novel and customer centric digital banking. The objectives are broadly categorized into specific goals and operational objectives to ensure clarity and alignment throughout the research and implementation process.

1.4.1 Project Goals

A goal is a general, long run desired state or situation that expresses the overall purpose of the project work. It should give direction, focusing on the question of "What to Achieve?". In general, the goals are qualitative and describe what will be the ultimate benefit of this venture.

This, in the context of the Araliya Bank project, means improving customer experience and satisfaction through effective deployment of AI technologies in the online transaction platform. Goal represents big picture outcomes that the project is supposed to achieve, such as improved operational efficiency, increased trust in the use of digital services, and a leading competitive edge.

- **Enhancing Customer Experience**

A goal is a broad objective or broad aspiration of a project, which defines what is ultimately aims to achieve. The primary objective of this project is to improve customer interaction and satisfaction through Araliya Bank's AI powered online transaction platform. This includes leveraging personalized recommendations, ensuring efficient transaction processing, and providing 24/7 accessibility. By focusing on such outcomes, this project will fulfill various customer needs and improve overall satisfaction and engagement with the bank's digital services. The goals provide a direction and vision to undertake the project. They represent the ground on which more concrete and actionable objectives are built.

- **Improving Operational Efficiency**

The project aims to find out how AI can contribute to automating banks' repetitive tasks to cut down processing time and preserve resources for better delivery of services. By streamlining processes, AI not only helps improve operational efficiencies but also ensures that customers' experience increases the pace and reliability of banking services. The project also tends to analyze the cost saving measures that AI technologies permit for assessing their impact on the overall performance of the bank in providing more affordable and efficient services to customers.

- **Ensuring Customer Trust and Security**

The project will assess the effectiveness of Araliya Bank's AI powered platform in securing customer data and transactions with the use of advanced mechanisms like fraud detection and biometric authentication. With the integration of AI based tools, the platform shall guarantee customers protection against unauthorized access and fraudulent acts that would eventually bring more trust and reliability in the use of digital banking services. The project will also map potential risks from the adoption of AI, such as data privacy issues, and suggest strategies for mitigating such challenges while adhering to regulatory standards and maintaining customer trust.

- **Promoting Digital Adoption**

The project will also test the usability of Araliya Bank's AI powered platform, testing for all kinds of customers, but focusing on those who are less familiar with technology. The research study will investigate problems that might be encountered by such users, which may indicate issues in the interfaces that make the technology difficult to use or may lack any guidance. Based on the findings, recommendations to implement better customer engagement, smoothing the onboarding processes, and assuring that the platform is inclusive to all users will be produced in this project.

- **Delivering Strategic Insights**

It also aims to provide holistic insights into strengths and limitations of Araliya Bank's AI powered online transaction platform and based on the successes and challenges of the implementations, present actionable recommendations to further improve the functionality, usability, and effectiveness of the platform. These will definitely help in further fine tuning the platform to meet customer needs and ensure that Araliya Bank maintains its competitive edge in the rapidly changing digital banking environment.

1.4.2 Project Objectives

The objective is an action that is specific, measurable, and can be taken towards the realization of a goal. Objectives answer the question "How will we achieve the goal?"> They are narrower in scope than goals, often focusing on individual tasks, processes, or outcomes.

e.g., For this project, objectives will include analyzing the features of the platform, analyzing customer satisfaction, investigating AI applications, and proposing ways of improvement.

These are clear-cut actions which contribute directly to fulfilling the larger goals.

- **Evaluation of Platform Features**

It will be conducted by undertaking a comprehensive analysis of the features provided by Araliya Bank's AI powered platform on personalization, speed of transaction, chatbot support, and multi-channel integration. From the analysis of such functionalities, the study shall establish their efficacy in providing a seamless and superior customer experience. The exact degree to which such features meet the needs and expectations of customers will be highlighted together with areas that need improvement that will enhance the overall usability and satisfaction with the platform.

- **Assessment of Customer Satisfaction**

The research in the following tends to collect important feedback from customers to check the satisfaction level of customers about Araliya Bank's AI powered platform. Through this, the research will find the various factors driving high and low user experience by collecting

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customer opinions. This feedback will let them understand the perception of the platform with regard to positive and negative areas, which will help in improving overall customer satisfaction and ensuring a better user experience.

- **Investigation of AI Applications**

The study shall delve into how AI technologies are integrated into various facets of Araliya Bank's platform for personalized financial advice, detection of fraudulent activities, and real-time customer support. Based on the study of the effectiveness of these AI-driven features, it will point out the areas where these functionalities can be further enhanced or expanded to provide more accurate recommendations, improve fraud detection capabilities, and offer more efficient support. The aim is to optimize the AI capabilities of the platform for a more personalized and secure experience for users.

- **Analysis of Technical and Operational Challenges**

This research project is also aimed at finding the analysis of any technical and operational issues that may affect the AI-driven platform of Araliya Bank. In particular, this is designed to research and analyze how failed performance, e.g., Could be an outage inaccuracy or slow processing will affect the results of the said platform. In addition, operational issues about customer resistance towards the adoption of AI technologies and lack of awareness about its features will be studied. The difficulties being understood will help determine where improvements should be made to have smoother users and broader acceptance of the platform.

- **Data Privacy and Security Review**

The project will also test Araliya Bank's AI-driven platform for compliance with local and international data protection regulations in order to ensure that customer data is handled according to the standards of the industry. It will also assess the adequacy of the existing security measures, including encryption and access controls, in place to protect customer information from unauthorized access. This will determine whether the platform has implemented a safe place for its users, with areas of possible improvement needed in security protocols in order not to lose customers and breach any regulatory compliance.

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- **Proposing Improvements**

It looks to cater, in particular, to the usability aspects through substantive recommendations that detail limitations within Araliya Bank's AI-powered platform. Conclusions on key areas where the interface can be leveraged are made to assist in smoothing out the user interface, enhancing the security environment, and ensuring quick and efficient access. Additionally, the project will provide recommendations on how to increase customer engagement in, and wider adoption of, AI-powered services by offering clearer educational resources, smoothing onboarding processes, and showing added value to the users of the platform in a more personalized way. Such recommendations will aim at further making the experience more user-friendly and secure while contributing to the growth and adoption of the platform.

- **Benchmarking and Industry Comparison**

The project will try to benchmark Araliya Bank's AI-powered platform against the most similar offering of competitors within the digital banking industry, underlining best practices and areas of differentiation. From the relative strengths and weaknesses entailed by different competing platforms, the study will point to ways in which Araliya Bank may utilize its distinctive features such as customer security protocols to outcompete others. In other words, this would determine points where the bank should leverage its strong points to meet not only the industry standards but also the expectations of customers so that Araliya Bank is in fast-growing digital banking environment.

These goals and objectives, when achieved, would give Araliya Bank a clear roadmap on how AI can help leverage its digital platform in improving customer experiences while remaining competitive and leading in the financial services industry.

1.5 Project Scope

The scope of the project for Araliya Bank Digital Banking Transformation will involve implementation, enhancement, and optimization of the bank's AI-powered online transaction platform to achieve better customer experience and satisfaction. The project boundaries will include key phases of development, integration, security features, and customer education, while clearly excluding areas not directly related to this digital transformation program.

1.5.1 Project Scope Boundaries

In-Scope,

- **AI-Powered Transaction Platform Development**
 - » Develop an AI-based system that permits online transactions by supporting real-time fraud detection, AI-powered customer support, such as through chatbots, and predictive financial advice.
 - » Integration with the AI platform into already running digital banking channels, mobile application, website, and ATM services.
- **Enhanced Customer Authentication**
 - » The implementation of secure authentication features, such as biometric recognition-face or fingerprint- and behavior analytics for secure and seamless transactions.
- **Mobile and Web Applications Enhancements**
 - » Reimagining Araliya Bank's mobile application and web platform into an intuitive and seamless UI/UX experience.
 - » Integrating the AI-powered system to offer a personalized experience in user engagement, based on transaction history, preferences, and financial behavior.

- **Data Analytics for Personalization**

- » AI and machine learning algorithms can be utilized for analyzing customer transaction data and giving personalized recommendations of offers and services.
- » Deploy appropriate AI models for the automation of product recommendations, financial insights, and customer engagement strategies.

- **Customer Support Automation**

- » AI-powered supported tool development and deployment, including virtual assistant and chatbots, to answer frequently asked questions, troubleshoot problems, and guide customers in real time.

- **Security and Fraud Detection**

- » Artificial Intelligence based fraud detection systems use machine learning to detect suspicious transactions, ensuring safe interaction in digital banking.
- » Implementation of encryption protocols, multi-factor authentication, and other measures to enhance security in digital banking.

- **Customer Education and Adoption**

- » Tutorials, webinars, and in-branch sessions would be initiated, which would train less tech-savvy customers for better use of the facilities given by digital banking.
- » Ensure easy adoption through FAQs, guides, and customer support concerning new features.

Out-of-Scop,

- **Core Banking System Overhaul**

- » This project does not aim at modifying or upgrading the core banking systems of Araliya Bank, which deal with internal banking processes such as account management and backend processing of transactions.

- **Physical Branch Upgrades**

- » This will be enhanced by means of a digital platform, however, this project will not include the transformation of the physical branch footprint such as branch redesigns or expansion.

- **Non-Digital Banking Channels**

- » The project does not cover non-digital channels, such as traditional telephone banking or in-person interactions, unless these are used as part of a hybrid system for AI enhanced support

- **Third-Party Service Integration**

- » It does not include integration with third-party services providers or external platforms, such as non-bank financial services, unless directly related to AI-powered solutions already include in the core services of the bank.

- **Full-fledged AI Model Training**

- » Although AI-based features will be included, the scope does not include developing complicated neural networks for specialized tasks beyond transaction-related suggestions or training new AI model from start on raw, unstructured data.

1.5.2 Project Deliverables

The Araliya Bank digital banking transaction will provide a complete online-based AI-powered transaction platform, which includes fraud detection in real time, secure authentication, and personalized advice about finance. This also includes the proposal for the optimization of the Araliya Bank mobile app and web, integrating AI-driven features for enhanced user experience. AI-powered customer support, such as chatbots and virtual assistants, will be implemented to provide real-time support and query resolution. In addition, it will implement robust security features such as AI-powered fraud prevention tools, biometric authentication, and encrypted channels. Finally, it will also roll out a major customer education program to facilitate seamless adoption of these advanced digital banking services.

1.6 Risks and Benefits of exploring the Practical applications and limitations of the use of AI to the Araliya Bank

1.6.1 Benefits

- **Enhancing Customer Experience**

AI in Araliya Bank can very well enhance customer experience through its personalized services and available 24/7. Analyzing customer data, AI servers banking experiences tailored to customers, suggesting products and services based on individual preferences and needs. In addition, AI-driven chatbots and virtual assistant operate continuously, enabling them to conduct banking inquiries and transactions any time of day without delay, enhancing access and facilitating ease. This combination of personalized interactions helps create a seamless, customer-centered banking environment.

- **Increased Operational Efficiency**

AI will contribute much to enhancing operational efficiencies in Araliya Bank by automating more routine tasks, such as data entry, transaction processing, and account management. The human resource thus freed can then be used on higher-order, value-added tasks. In addition, AI helps speed up decision making since it rapidly sorts out volumes of data and is in a position to provide quicker and better decisions on various matter related to

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leading, fraud detection, and risk assessment. This triggers better overall productivity, faster customer service, and more precise financial analysis.

- **Cost Reduction**

AI can support Araliya Bank in shrinking the cost associated with maintaining operational processes because chatbots and automated systems have the potential to replace huge groups of customer support people. AI will provide optimal resource utilization: it is effective in handling both human and technological resources; therefore, Araliya Bank may get streamlined into operational process by providing substantial cost savings. With better efficiency, the bank would be in a position to utilize better the available resources for operational tasks.

- **Advanced Fraud Detection and Risk Management**

AI will enhance fraud detection and risk management in Araliya Bank by analyzing transaction patterns in real time for suspicious Activities that may reduce the fraud risk. Predictive analytics of a AI can also predict the occurrence of any potential risk, such as loan defaults or financial crises, thus enabling the bank to take prior measures to mitigate such risks. The early detection of fraud and anticipation of future challenges enhance the bank's security and decision-making for the protection of both the bank and its customers.

- **Innovation and Competitiveness Edge**

It positions Araliya Bank as a technologically leading bank for digital banking and therefore would be more attractive for the tech-savvy customers who consider innovation and rich advanced services. AI paves the way for new business models, allowing the development of innovative financial products such as AI-driven investment tools, tailored savings plans, and personalized insurance policies. All these will bring value not only to the customers by ensuring customer satisfaction but also to the bank in offering differential advantages among fast evolving competition in the financial industry.

1.6.2 Risks and Limitations

- **Data Privacy and Security Concerns**

With AI in Araliya Bank, voluminous amounts of customer data have to be collected and analyzed; thus, this calls for privacy and security concerns. They may have queries about their information storage and utilization, particularly about compliance with the data protection laws. In addition, AI systems tend to become highly attractive targets for cyberattacks; compromise could mean heavy data breaches and financial loss. Addressing these vulnerabilities becomes crucial. Addressing these vulnerabilities becomes crucial to maintain trust in the sale of sensitive information.

- **High Initial Investment and Maintenance Costs**

This would require huge upfront investments in infrastructure, training and development of the AI models. These may not be easy on the financial and operational resources of Araliya Bank. Secondly, ongoing costs are involved with AI systems to keep them updated, monitored, and fine-tuned so that they continue to be accurate and efficient. While high, these costs are necessary for full realization and sustenance of benefits from AI in the long run.

- **Customer Trust and Acceptance**

Some customers, especially the elderly, might resist AI-driven banking services and instead prefer traditional methods where there is one-on-one interaction. The overdependence on AI in banking will deprive the sector of the human touch, making it appear impersonal or too automated. This may impact customer satisfaction since most people still believe in direct communication and personalized attention by human representatives for their financial needs.

- **Risk of Bias and Inequality**

AI systems can inadvertently perpetuate biases in the training data, leading to unfair or discriminatory decisions on loan approvals or credit scoring. These biases can disadvantage certain groups, especially those whose behaviors or financial patterns run contrary to the

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norm recognized by the AI. This exclusion from personalized services or financial opportunities highlights the need for ongoing monitoring and refinement of AI algorithms to ensure fairness and inclusiveness in their application.

- **System Failures and Reliability Issues**

No AI system in banking is immune to operational malfunctioning because of problems that might arise either at the data processing stage or include some algorithm inaccuracies, disrupting services or their results and jeopardizing customer confidence. Besides that, some really complicated AI models practically lack transparency necessary for explaining justifications behind individual decisions. This opacity creates a lot of skepticism among customers and regulators, hence strong systems and appropriate communication go a long way in the upkeep of reliability and trustworthiness.

1.6.3 Mitigating Risks and Maximizing Benefits

While AI can offer immense capabilities, Araliya Bank will have to seek strong data governance practices that not only make sure all sorts of regulations are complied with but also protect customer information. Human interaction will need to balance AI-driven services, in order to build and manage trust, and also addressing myriad customer needs. These will be continuously reviewed and tasted so as to find biases, inaccuracies, and operational issues that have to be fixed. Moreover, education customers about AI-driven services and providing clear guidance during transitions can foster acceptance and confidence in the technology.

1.7 Key Deliverables of the Project

- **Comprehensive Project Management Plan**

A comprehensive plan that outlines the goals, scope, stakeholders, deadlines, deliverables, and success criteria of a project. This document acts as a guide to guarantee efficient use of resources and conformity with organizational goals.

- **Research Report**

A report that summarizes findings from focus groups, industry benchmarking, and consumer surveys. It draws attention to the competitive environment, consumer preferences, and important areas for development.

- **Action Plan**

A detailed implementation strategy for the AI-powered platform upgrades. To accomplish the project's goals, it outlines each activity's duties, roles, responsibilities, and deadlines.

- **Business Case**

An explanation document outlining the anticipated advantages of the AI-powered platform, such as increased financial return on investment, operational effectiveness, and customer happiness. It also lists other choices, and the trade-offs associated with them.

- **Budgetary Report**

A comprehensive budget that accounts for platform updates, system upkeep, advertising, employee training, and emergency funding.

- **Risk Analysis and Management Plan**

Identifying possible hazards, such as system outages, data breaches, and unhappy customers. The plan includes contingency plans, monitoring systems, and mitigation techniques.

- **Work Breakdown Structure (WBS)**

A hierarchical task breakdown that divides the project into digestible parts for resource allocation and progress monitoring.

- **Time Management Plan (Gantt Chart)**

A graphic timeline that displays tasks, dependencies, deadlines, and project milestones. It guarantees efficient time management and on-time outputs.

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- **AI Integration Framework**

The creation and use of scalable infrastructure to incorporate artificial intelligence into the current internet marketplace. Data pipelines, algorithm deployment techniques, and APIs are all included.

- **Customer Experience Enhancement Tools**

Creation of tools to enhance customer interaction and happiness, such as transaction suggestions engines, AI-driven chatbots, and customized dashboards.

- **Data Security and Compliance Protocols**

Use of strong access controls, cutting-edge encryption, and adherence to local and global data protection laws as well as legislation like the GDPR. This fosters confidence and guarantee the integrity of client data.

- **Predictive Analytics Models**

Creation of AI models to foresee client requirements, identify fraudulent activity, and offer proactive assistance.

- **Training and Adopting Plans**

Training courses for employees and customer onboarding materials (tutorials, FAQs, and videos) to encourage platform uptake and effective use.

- **Improved Reporting and Dashboard Tools**

Dashboard and real-time reporting tools for tracking KPIs, system performance, and customer interaction allow for data-driven decision making.

- **Customer Feedback Mechanism**

Use of feedback loops collect consumer information for ongoing development, such as surveys, Net Promoter Scores (NPS), and sentiment analysis via chatbots.

- **Documentation and Reporting**

For future reference and scalability, comprehensive documentation of all system setting, training materials, user manuals, and project milestones should be kept.

- **Innovation Service Offerings**

Creation of fresh AI-powered services to set Araliya Bank apart from rivals, such tools for financial planning and loyalty schemes.

1.8 Assumption, Constraints and Limitations

1.8.1 Assumption

- **Customer Adoption of Technology**

Customers are presumed to be eager to use AI-powered platforms and to have a basic understanding of digital banking services.

- **AI Training Data Quality**

The AI system will have access to enough high-quality data to efficiently train its models, guaranteeing precise suggestions and predictions for transactions.

- **Reliable Internet Connectivity**

Customers are presumed to have reliable internet connections in order to make effective use of the platform.

- **Security Compliance**

Local and international data security requirements, such as encryption and privacy laws (such as the GDPR and Sri Lanka's Data Protection Act), would be adhered to by the AI-powered platform.

- **Continual Improvement of AI Models**

In order to maintain relevance and accuracy over time, it is anticipated that the AI algorithm would be modified often in response to fresh data and user input.

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- **Sufficient Infrastructure**

The platform is predicated on a strong IT infrastructure that can sustain AI features and real-time transaction processing without experiencing prolonged outages.

1.8.2 Constraints

- **Regulatory Constraints**

The types of data that may be handled or the methods in which AI can be used in financial transactions may be restricted by local financial legislation and compliance requirements.

- **Technology Limitations**

The computing capacity of AI models may be constrained, which might cause processing delays, particularly for intricate transactions or real-time analysis.

- **Customer Demographics**

Non-techies who have trouble with digital banking may make up a sizable section of the clientele, which would reduce the platform's usefulness for them.

- **Data Privacy**

Limitations on data usage may be necessary to ensure that AI models respect client privacy, which would reduce the breadth of insights that AI may extract.

- **Platform Interoperability**

The capacity of the AI-powered system to smoothly integrate with all other financial systems, especially legacy systems, may be limited.

- **Budget and Resource Constraints**

Budget constraints may affect the deployment of state-of-the-art AI algorithms, security protocols, or the platform's capacity to grow efficiently.

1.8.3 Limitations

- **Limited Handling of Complex Queries**

Even while AI can effectively answer simple questions, complicated or extremely subtle problems would still need human interaction, which would restrict the platform's ability to meet all client requests.

- **AI Decision Making Errors**

AI algorithms may produce inaccurate forecasts or suggestions, particularly when training data is inadequate or skewed.

- **Over Reliance on Technology**

If there are technical problems with the AI-powered platform, including system outages, or if the AI answers are incorrect, customers can become frustrated.

- **Fraud Detection Challenges**

Customers may experience annoyance when normal transactions are flagged as fraudulent by AI algorithms used for fraud detection. On the other hand, if they are not updated often, they may overlook new fraud trends.

- **Initial Learning Curve**

Customers may need to devote time to learning how to utilize the platform efficiently, especially if they are elderly or less tech-savvy.

- **Cultural and Behavioral Diversity**

AI systems could find it difficult to comprehend the entire spectrum of linguistic, behavioral, and cultural variety among consumers, which could result in inaccurate product suggestions or customer profiling.

2. Project Human Resource Management

2.1. Introduction

In this regard, project human resource management plays an important role in using effective human resources involved with correct application and well-defined roles and responsibilities within the working team. From this perspective, the project undertaken for Araliya Bank's AI-powered online transaction platform will involve all aspects, right from planning, development, implementation, and frequent maintenance, continuously. Human resources management will help in ensuring smooth coordination, high efficiency, and success in meeting project goals for improving customer experience and satisfaction.

The project will involve various departments within the bank including third-party vendors and technical teams in designing, deploying the AI-powered platform. Key success factors will be derived from the internal team structure, skill sets, and clear definitions of roles for the effective delivery of an enriched banking experience on the platform.

2.2. Project Team / Team Hierarchy

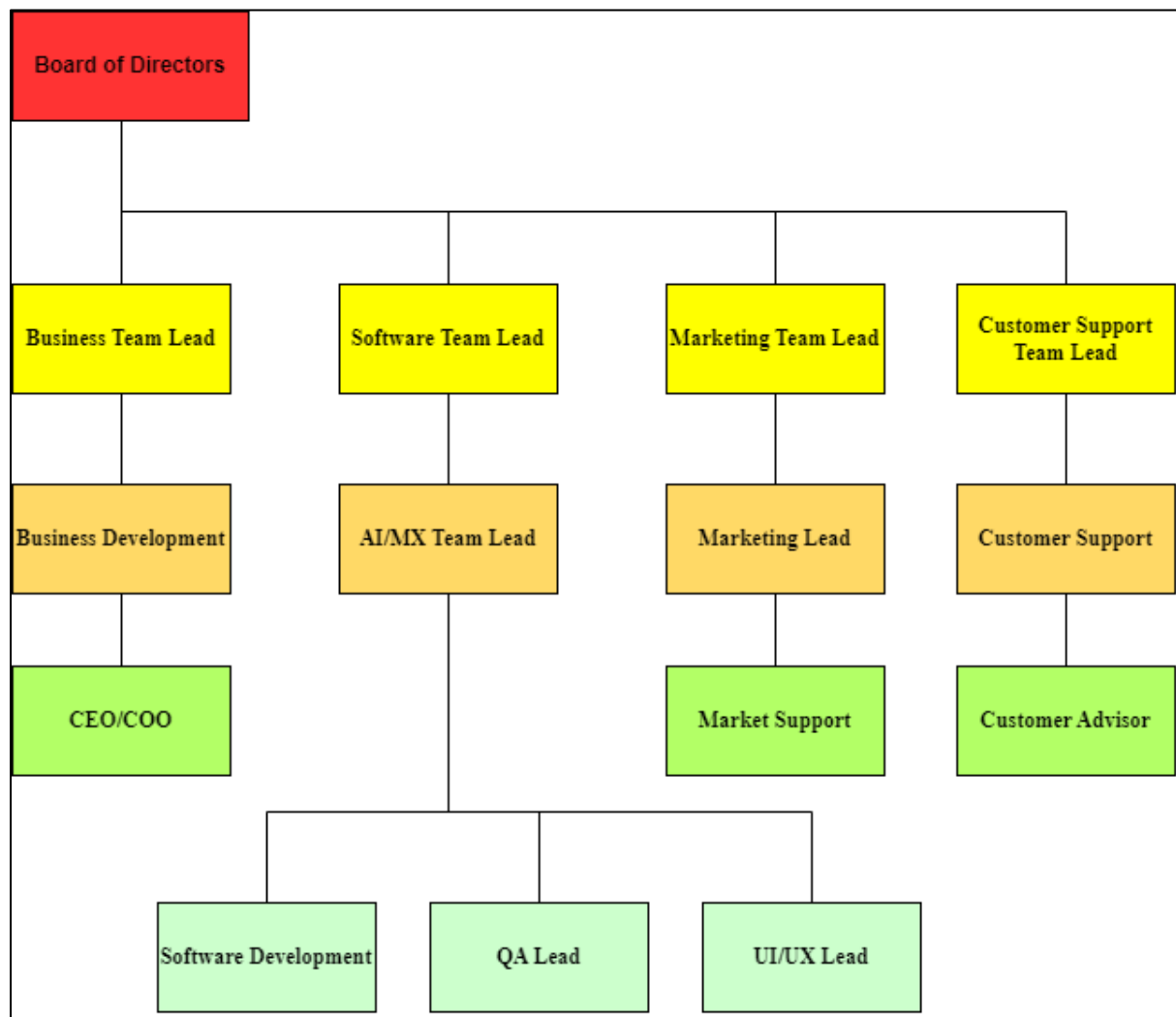


Figure 1 – Team Hierarchy

2.3.Roles & Responsibilities of the Project Team

- **Board of Directors**

The Board Directors shall assume responsibility for providing strategic guidance and oversight over the AI-powered transaction platform project at Araliya Bank. They are responsible for approving major project decisions, including budgets, and ensuring that the platform aligns with the broader organizational goals and policies of the bank. The Board makes sure that high-level guidelines is provided with an assurance that resources will be put to good use in the successful delivery of the project in keeping with the Bank’s vision and long-term objectives.

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- **Business Team Lead**

Role -: The business team lead is responsible for overseeing the business strategy of the AI-powered platform to ensure alignment with customer needs and the bank's organizational goals. This includes making sure the platform supports key business objectives such as customer acquisition, retention, and profitability. The Business Team Lead work with different departments for the project to meet business objectives and manages the resource allocation in the right manner so that timelines can be adhered to and drive the project forward efficiently.

Responsibilities -:

- » Lead the development and management of the business roadmap of AI-powered platforms.
- » Liaise with software and marketing teams to align technology with business requirements.
- » Monitoring business metrics to assess the success of the platform in meeting customer needs.
- » Pinpoint new business opportunity based on customer insight.

- **Software Team Lead**

Role -: Software shall be engaged in the management of development activities with technical implementation, which means taking care that functional, technical, and security requirements of the software are met. He will interact closely with the AI/ML team, QA lead, and UI/UX lead to develop and deliver seamless, robust solutions with a go to market objective function of the platform. The software team lead will ensure the platform is developed in an effective manner, assuring high standards of quality with all technical components integrated into the best possible user experience.

Responsibilities -:

- » Lead the development and deployment of the platform by a team of software developers.
- » Ensure the technical feasibility and scaling of the platform.
- » Review and approve technical designs, code, and architecture.
- » Troubleshooting and resolving technical challenges during the development and maintenance phases.

• **Market Team Lead**

Role -: The marketing team lead shall spearhead the marketing campaign towards ensuring the AI-powered transaction platform meets the needs of customers by properly and efficiently communicating its features, benefits and value propositions. This covers developing a strategy to be adopted for customer engagement and the creation of awareness for the adoption use of the benefits accruable from the platform, Through joint efforts, the market team lead ensures that the marketing efforts meet the objectives set out for the business and reach out to the right audience in pursuit of its growth and success on the platform.

Responsibilities -:

- » Design marketing campaigns to increase awareness and usage of the platform.
- » Collaborate with the business team in assuring that marketing efforts are pointed toward business objectives.
- » Conduct market research to identify customer needs, preferences, and pain points.

- » Measure and report on the effectiveness of marketing campaigns to adjust strategies appropriately.

- **Customer Support Team Lead**

Role -: The customer support team lead will be responsible for leading the customer support operations of the AI-powered transaction platform, ensuring that the team provides premium assistance to the users. This will involve ensuring seamless onboarding of customers, solving issues in the shortest time possible, and follow-through for continued improvement of the user experience. This also includes handling escalation in an effective manner and ensuring customer satisfaction by the quick and efficient resolution of complex problems, hence keeping the service level at a high point in customer journey view.

Responsibilities -:

- » Oversee the operations of customer support, including training for support staff.
- » Lead a customer support team in the development of strategies that will enhance the customer experience.
- » Analyze customer feedback for opportunities to improve the platform.
- » Ensure timely and effective resolutions of AI-platform related support issues.

- **Business Development Team**

Role -: The business development team shall be tasked with the identification and development of opportunities regarding the bank's AI-powered transaction platform expansion, through strategic partnerships or new customer acquisition. This includes exploring new markets, building relationships with potential partners, and developing initiatives to attract more users. While onboarding the customer base and it grew in strength, it made sure the platform showed strong ROI; the platform was adopted and grew overall revenue to meet the bank's long term business goals.

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Responsibilities -:

- » Market research and identification of new business opportunities
- » Partnership-building and collaboration for expanding the platform's research.
- » Work with the business team for product and services optimization

• **AI/MX Team Lead**

Role -: The AI/MX team lead will be responsible for oversight in the development, deployment, and continuous improvement of models for AI in the platform to meet business requirements. This includes aligning the solution of AI for customer needs and business objectives, driving personalization, and overall improvement of functionality. The AI/MX team lead works closely with other technical teams to embed AI smoothly into the platform, ensuring it enhances the user experience, meets business goals, and continues to advance the capabilities of the platform.

Responsibilities -:

- » Lead the AI and MX team in the development of transaction processing, fraud detection, and personalization algorithms.
- » Liaise with the software team for the integration of AI models into the platform.
- » Monitor the performance of AI, making changes and enhancements when necessary to make the user's experience optimal.
- » Ensure that solutions proposed AI are ethical and follow regulations concerning data privacy.

AI -: Artificial intelligence

MX -: Machine learning

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- **Software Development Team**

Role -: The software development team shall design, develop, and implement the technical components of the AI-powered transaction platform. This shall involve very close collaboration with the AI/MX, QA, and UL/UX teams to ensure the platform is integrated, functional, and high performing. Development of new features, coding, testing, resolving technical problems, and integration of all components to ensure that the user interface works seamlessly is what the team is focused on, keeping scalability and security considerations across the platform.

Responsibilities -:

- » Oversee the team of developers that will create and implement the platform
- » Verify the platform's scalability and technological viability.
- » Examine and accept the architecture, code, and technical designs.
- » During the development and maintenance stage, troubleshoot and fix technological issues

- **QA Lead**

Role -: The QA Lead will oversee the quality assurance activities to ensure a bug free, reliable, fully optimized AI-powered transaction platform. In this regard, the lead is supposed to manage all comprehensive testing across the different modules of functionality, performance, security, etc., of the said platform for conformance to standards. The QA lead will be involved with the software development team in finding and resolving issues to make the platform provide a seamless user experience and work under all conditions.

Responsibilities -:

- » Create and run test cases to verify the platform's performance and functionality.

- » Verify that every feature and improvements satisfy the platform's quality requirements.
- » During development, find problems and work with the software team to fix them
- » Assure platform performance and scalability across a range of load scenarios

- **Marketing Lead**

Roel -: Concentrate on putting platform marketing efforts into action and boosting the AI-powered transaction system's brand awareness.

Responsibilities -:

- » Manage and carry out marketing activities, including social media campaigns, content production, and internet advertising
- » Track performance indicators to enhance client acquisition and maximize marketing efforts
- » Work together with other marketing departments to match business objectives with campaigns

- **Market Support**

Role -: Adapt tactics based on real-time consumer contracts and offer on the ground assistance for market movements and client feedback.

Responsibilities -:

- » Surveys and consumer input should be gathered to guide future platform enhancements.
- » To make sure the platform is competitive, help with the analysis of rivals and market trends.

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- » Collaborate with the business team to identify client problems and make suggestions for solutions.

- **Customer Advisor**

Role -: Give clients tailored advice to help them use the platform to its fullest potential

Responsibilities -:

- » Offer customers personalized advice and recommendations about their transaction history and preferences
- » Assist customers in troubleshooting problems they experience in using the platform
- » Communicate and escalate major concerns to the customer support team and the business team

- **CEO/COO**

Role -: The CEO/COO will oversee the entire AI-powered transaction platform project for alignment with the strategic vision and operational goals of Araliya Bank, ranging from taking ultimate responsibility in ensuring the success of the platform right from its implementation to customer adoption. The CEO/COO makes high-level decisions on resource allocation, project direction, and timelines to ensure that the platform meets the ensure that the platform meets the business and customer needs while delivering value to the bank.

Responsibilities -:

- » Make high-level strategic decisions and ensure proper resources allocation within departments.
- » Liaise with the board of directors in aligning the project to broader bank objectives.
- » See how the AI-powered platform can help in the business of the bank or impact their customers' performance.

3. Project Scope Management

3.1. Introduction

Project Scope Management is one of the most important aspects of project management, relating to defining, controlling, and monitoring all the work required to deliver a successful project. In implementing and enhancing its AI-powered online transaction platform, thereby improving customer experience and satisfaction, effective scope management guarantees that all the objectives of the different projects are in line with that goal.

The clear definition of scope involves defining the deliverable, boundaries, and objectives of the project. This involves the development of a detailed Work Breakdown Structure to be used to outline the tasks at hand for the successful implementation of such projects. The scope also involves monitoring and controlling in terms of preventing scope creep so as to keep the project on target regarding time, budget, and quality. Robust scope management should be incorporated into the project for a number of reasons, it aligns the outcomes of the project with the needs of the customer, it prevents extra work that might delay the implementation, and it assures efficient use of resources. With clearly defined deliverables, the bank can fully realize the potential of its AI platform to offer personalized banking solutions that will increase customer satisfaction and lead to long-term trust.

Effective scope management eventually drives the successful delivery of the project, ensuring that all stakeholders are satisfied with the outcomes while maintaining the overall strategic vision of improving digital banking services.

3.2. Scope Inclusions

The scope of this project is to increase customer pleasure and experience by implementing and improving Araliya Bank's AI-powered online transaction platform. The scope's primary inclusions are,

- **Development and Deployment of AI Features**

The project involves the integration of various advanced AI algorithms to offer personalized banking experience to customers based on their needs. This also involves the use of intelligent chatbots and virtual assistant for continuous, 24/7 customer support for better

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accessibility and responsiveness. Mechanisms for fraud detection and prevention using AI will also be deployed to ensure secure and reliable transactions, thereby instilling greater trust and confidence among customers in the platform. Together, these innovations are designed to elevate the overall customer experience and satisfaction.

- **User Interface (UI) and User Experience (UX) Enhancements**

It designs and interface that it easy to use and ensures navigation is easy; hence, customers can easily access the services of the bank. Secondly, it ensures ease of access on both mobile and web platforms for easy access and use that enables the smooth and efficient operation of digital banking across different devices and user preferences.

- **Customer Feedback Integration**

It also entails the implementation of features that can help in the collection and analysis of customer feedback for the continuous improvement of the platform. The project will utilize AI tools to find trends and patterns in customer preferences and behavior such that the platform will continuously evolve to meet user needs effectively, enhancing overall customer satisfaction.

- **Transaction Processing Improvement**

The project will automate transaction workflows to reduce processing time, hence making it easier and more efficient for customers. It also enhances real-time transaction tracking and notification systems, ensuring users are always up to date with their banking activities in a timely and secure manner.

- **Security and Compliance**

The project focuses on enhancing data security by implementing AI-driven encryption and state of the art monitoring tools to protect sensitive customer data. Compliance with local and international banking regulations also build trust and reliability in the platform, thereby upholding industry standards.

- **Employee Training and Support**

The project also involves comprehensive training sessions for the staff so that they are able to manage and maintain the AI-powered platform. A strong support framework will be developed for technical troubleshooting and system updates to ensure that the platform works seamlessly and with increased efficiency over time.

- **Performance Training and Support**

The project encompasses the integration of AI tools to ensure real-time performance monitoring of the platform for timely identification and resolution of issues. In addition, it involves the development of intuitive dashboards for tracking KPIs that provide actionable insights in optimizing the functionality of the platform for overall customer experience improvement.

3.3. Scope Exclusions

Although the project's goal is to develop Araliya Bank's AI-powered online transaction platform in order to increase customer pleasure and experience, several features and activities are outside its stated parameters. The following are excluded,

- **Hardware Procurement and Infrastructure Setup**

The project does not include procurement or physical deployment of servers, network devices, or other hardware components, it assumes that the current infrastructure will be adequate to support enhancements to the platform planned for.

- **Third-Party Software Development**

This also involves the development and customization of unrelated third-party applications or systems that will not relate to the AI-powered platform but form part of an integrated environment. It focuses solely on the integration of pre-approved and supported software solutions, ensuring seamless functionality and compatibility within the existing ecosystem.

- **Legacy System Upgrades**

The project scope does not include updating or replacing older systems that are not directly connected to the AI transaction platform.

- **Non-Ai Banking Upgrades**

Exclusion apply to improvements made to non-digital financial services like ATMs or in-branch procedures.

- **Physical Customer Interaction Channels**

The scope does not include enhancements to physical consumer engagement channels like call centers or physical branches.

- **Third – Party Parter Integration**

Integration with third-party APIs, e-commerce sites, or external payment gateways that were not previously enabled by the bank’s ecosystem is not included.

- **End-user Device Support**

It excludes direct troubleshooting or assistance for client devices, such as laptop and cell phones. Platform-side improvements continue to be the key emphasis.

- **Post-Implementation Maintenance**

After the project is finished long-term maintenance and support services are not included. This will be covered is a different contract

- **Marketing and Customer Awareness Campaigns**

Excluded are the development and implementation of marketing plans or consumer education initiatives pertaining to the new platform functionalities.

- **Regulatory Policy Changes**

Unless otherwise noted in a modification request, addressing or implementing policy changes beyond current rules during the project lifetime is not included

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3.4. Work Breakdown Structure (WBS)

The WBS for implementation and enhancing Araliya Bank's AI online transaction platform has been structured into hierarchical levels to ensure clarity and effective project management. A tabulated detailed WBS structure,

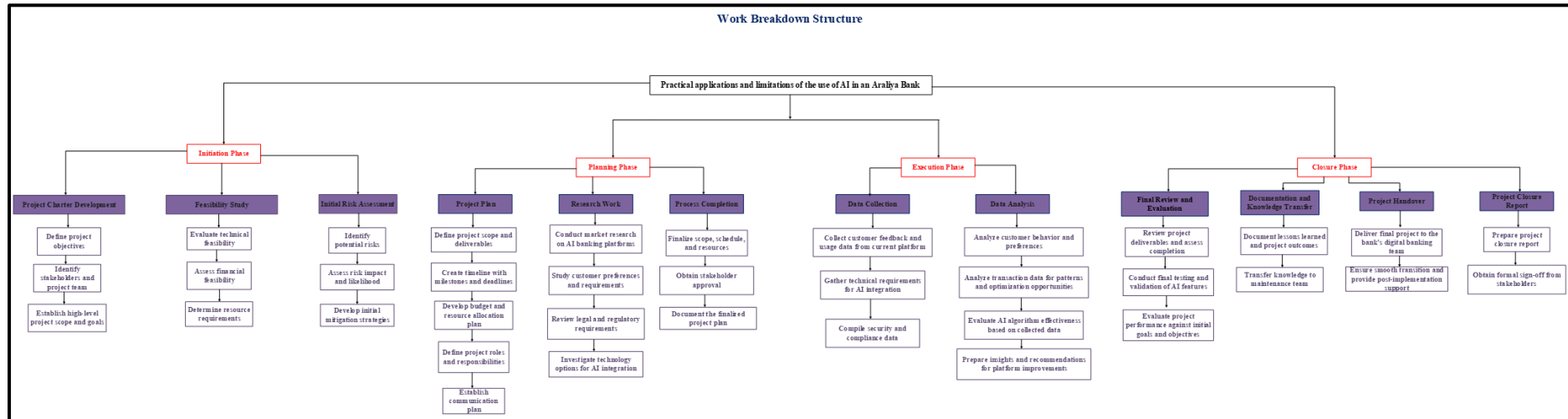


Figure 2 – Work Breakdown Structure

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Task ID	Tasks
01	Start
02	Initiation Phase
2.1	Project Charter Development
2.1.1	Define project objectives
2.1.2	Identify stakeholders and project team
2.1.3	Establish high-level project scope and goals
2.2	Feasibility Study
2.2.1	Evaluate technical feasibility
2.2.2	Assess financial feasibility
2.2.3	Determine resource requirements
2.3	Initial Risk Assessment
2.3.1	Identify potential risks
2.3.2	Assess risk impact and likelihood
2.3.3	Develop initial mitigation strategies
03	Planning Phase
3.1	Project Plan
3.1.1	Define project scope and deliverables
3.1.2	Create timeline with milestones and deadlines
3.1.3	Develop budget and resource allocation plan
3.1.4	Define project roles and responsibilities
3.1.5	Establish communication plan
3.2	Research Work
3.2.1	Conduct market research on AI banking platforms
3.2.2	Study customer preferences and requirements
3.2.3	Review legal and regulatory requirements
3.2.4	Investigate technology options for AI integration
3.3	Process Completion
3.3.1	Finalize scope, schedule, and resources
	Obtain stakeholder approval
	Document the finalized project plan
04	Execution Phase

Figure 3 – WBS (1)

4.1	Data Collection
4.1.1	Collect customer feedback and usage data from current platform
4.1.2	Gather technical requirements for AI integration
4.1.3	Compile security and compliance data
4.2	Data Analysis
4.2.1	Analyze customer behavior and preferences
4.2.2	Analyze transaction data for patterns and optimization opportunities
4.2.3	Evaluate AI algorithm effectiveness based on collected data
4.2.4	Prepare insights and recommendations for platform improvement
05	Closure Phase
5.1	Final Review and Evaluation
5.1.1	Review project deliverables and assess completion
5.1.2	Conduct final testing and validation of AI features
5.1.3	Evaluate project performance against initial goals and objectives
5.2	Documentation and Knowledge Transfer
5.2.1	Document lessons learned and project outcomes
5.2.2	Transfer knowledge to maintenance team
5.3	Project Handover
5.3.1	Deliver final project to the bank's digital banking team
5.3.2	Ensure smooth transition and provide post-implementation support
5.4	Project Closure Report
5.4.1	Prepare project closure report
5.4.2	Obtain formal sign-off from stakeholders
06	Finish

Figure 4 – WBS (2)

3.5. Work Breakdown Structure Dictionary (WBD)

WBS is an elaborate dictionary that describes what each WBS element is, what it will deliver, who is responsible, and other milestones associated with that element. Below is a key element of the Araliya Bank AI-powered online transaction platform project.

- **Initiation Phase**

- » **Project Charter Development**

- » **Objective -:** Documenting the project's objectives, stakeholders, and high-level scope will lay the groundwork for it.

- **Define project objectives -:** Determine and record the project's main objectives
- **Identify stakeholders and project team -:** Assemble the project team and determine the important stakeholders.
- **Establish high-level project scope and goals -:** Describe the project's goals and general scope

- » **Feasibility Study**

- » **Objective -:** Examine the project's feasibility from a technical, financial, and resource standpoint.

- **Evaluate technical feasibility -:** Assess if the project's AI integration can be accomplished with the technologies already in use.
- **Assess financial feasibility -:** Analyze the project's cost effectiveness and budgetary needs.
- **Determine resource requirements -:** Determine the resources, technology, and people required to finish the project.

» **Initial Risk Assessment**

» **Objective -:** Determine possible hazards and create preliminary plans for mitigation them.

- **Identify potential risks -:** Determine the external and internal threats to the project's success.

- **Assess risk impact and likelihood -:** Assess each risk's seriousness and likelihood of happening

- **Develop initial mitigation strategies -:** Create plans to reduce or control the hazards that have been identified.

• **Planning Phase**

» **Project Plan**

» **Objective -:** Create a through project plan that details responsibilities, resources, budgets, and schedules.

- **Determine resource requirements -:** Determine the project precise technology, equipment, and human resources needed for the project

- **Create timeline with milestones and deadlines -:** Create a project schedule that includes important dates and benchmarks.

- **Develop budget and resource allocations plan -:** Arrange the project's resources and budget allocations.

- **Define project roles and responsibilities -:** Assign roles and duties to each member of the project team.

- **Establish communication plan -:** Create a communication plan for upgrading stakeholders on the status of the project.

- » **Research Work**
- » **Objective -:** During the planning and design stages, carry out research to aid in decision-making.

- **Conduct market research on AI banking platforms -:** Examine the market's AI-powered banking platforms.

- **Study, customer preferences and requirements -:** Examine consumer preferences and requirements to inform platform design.

- **Review legal and regulatory requirements -:** Examine the platform's applicable legal and compliance requirements.

- **Investigate technology option for AI integration -:** Examine possible AI technologies that might be include into the platform.

- » **Process Completion**
- » **Objective -:** Complete and record the project plan for approval by the stakeholders.

- **Finalize scope, schedule, and resources -:** Verify the project's budget, timeline, and resources distribution.

- **Obtain stakeholder approval -:** Obtain stakeholders' official approval of the completed plan.

- **Document the finalized project plan -:** Compile and record the authorized project plan.

- **Execution Phase**

- » **Data Collection**

- » **Objective -:** Collect the information required for analysis and AI integration.

- **Collect customer feedback and usage data from current platform -:** Gather opinions and usage trend from the existing platform.
- **Gather technical requirements for AI integration -:** Determine the technological requirements needed to include AI elements.
- **Compile security and compliance data -:** To guarantee security and adherence to legal requirements, collect pertinent information.

- » **Data Analysis**

- » **Objective -:** Examine the gathered information to guide platform enhancement and AI integration.

- **Analyze customer behavior and preferences -:** Examine consumer information and comments to determine preferences.
- **Analyze transaction data for patterns and optimization opportunities -:** To make changes, find trends in transaction data.
- **Evaluate AI algorithms effectiveness based on collected data -:** Test and assess AI algorithms' performance.
- **Prepare insights and recommendations for platform improvements -:** Create useful insights based on data analysis to improve the platform.

- **Closure Phase**

- » **Final Review and Evaluation**

- » **Objective -:** To evaluate the project's progress and confirm AI features, do a final evaluation.

- **Review project deliverables and access completion -:** Examine project deliverables to make sure they fulfil the stated goals.
- **Conduct final testing and validation of AI features -:** To make sure the AI features function as intended, conduct final testing.
- **Evaluate project performance against initial goals and objectives -:** Evaluate the project's performance in relation to its initial objectives and success standards.

- » **Documentation and Knowledge Transfer**

- » **Objective -:** Keep track of the project's results and impart expertise to the maintenance crew.

- **Document lessons learned and project maintenance outcomes -:** Document insights on post project maintenance and lessons learnt.
- **Transfer knowledge to maintenance team -:** Assure a seamless transfer of knowledge to the group in charge of continuing platform upkeep.

» **Project Handover**

» **Objective -:** Five the project to the bank's digital banking staff officially.

- **Deliver final project to the bank's digital banking team -:** Give the operational team the finished platform.
- **Ensure smooth transition and provide post implementation support -:** Provide assistance to guarantee a seamless transfer and address and problems that may come up.

» **Project Closure Report**

» **Objective -:** Obtain official stakeholder permission and document the project's conclusion.

- **Prepare project closure report -:** Record the project's outcomes, difficulties, and successes.
- **Obtain formal sign-off from stakeholders -:** Obtain official stakeholder sign-off to verify the project's completion.

3.6. Milestone Chart

Task Name	Duration	Start	Finish	Predecessors
Initial Prase	10 days	2024/11/01	2024/11/10	
• Project Charter Development	3 days	2024/11/01	2024/11/03	
- Define project objectives	1 day	2024/11/01	2024/11/01	
- Identify stakeholders and project team	1 day	2024/11/02	2024/11/02	2
- Establish high-level project scope and goals	1 day	2024/11/03	2024/11/03	3

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• Feasibility Study	4 days	2024/11/04	2024/11/07	4
- Evaluate technical feasibility	2 days	2024/11/04	2024/11/05	
- Assess financial feasibility	1 day	2024/11/06	2024/11/06	5
- Determine resource requirements	1 day	2024/11/07	2024/11/07	6
• Initial Risk Assessment	3 days	2024/11/08	2024/11/10	7
- Identify potential risks	1 day	2024/11/08	2024/11/08	
- Assess risk potential risks	1 day	2024/11/09	2024/11/09	8
- Develop initial mitigation strategies	1 day	2024/11/10	2024/11/10	9
Planning Phase	16 days	2024/11/11	2024/11/25	10
• Project Plan	7 days	2024/11/11	2024/11/17	
- Determine resource requirements	1 day	2024/11/11	2024/11/11	
- Create timeline with milestones and deadlines	2 days	2024/11/12	2024/11/13	11
- Develop budget and resource allocation plan	2 days	2024/11/14	2024/11/15	12
- Define project roles and responsibilities	1 day	2024/11/16	2024/11/16	13
- Establish communication plan	1 day	2024/11/17	2024/11/17	
• Research Work	6 days	2024/11/18	2024/11/23	14
- Conduct market research on AI banking platforms	2 days	2024/11/18	2024/11/19	
- Study customer preferences and requirements	1 day	2024/11/19	2024/11/19	15
- Review legal and regulatory requirements	2 days	2024/11/20	2024/11/21	16

- Investigate technology options for AI integration	1 day	2024/11/22	2024/11/22	17
• Process Completion	3 days	2024/11/23	2024/11/25	18
- Finalize scope, schedule, and resources	1 day	2024/11/23	2024/11/23	
- Obtain stakeholder approval	1 day	2024/11/24	2024/11/24	
- Document the finalized project plan	1 day	2024/11/25	2024/11/25	19
Execution Phase	23 days	2024/11/26	2024/12/18	20
• Data Collection	10 days	2024/11/26	2024/12/05	
- Collect customer feedback and usage data from current platform	3 days	2024/11/26	2024/11/28	
- Gather technical requirements for AI integration	4 days	2024/11/29	2024/12/02	21
- Compile security and compliance data	3 days	2024/12/03	2024/12/05	22
• Data Analysis	13 days	2024/12/06	2024/12/18	23
- Analyze customer behavior and preferences	3 days	2024/12/06	2024/12/08	
- Analyze transaction data for patterns and optimization opportunities	3 days	2024/12/09	2024/12/11	
- Evaluate AI algorithm effectiveness based on collected data	4 days	2024/12/12	2024/12/15	24
- Prepare insights and recommendations for platform improvements	3 days	2024/12/16	2024/12/18	25

Closure Phase	13 days	2024/12/19	2024/12/28	26
Final Review and Evaluation	4 days	2024/12/19	2024/12/22	
- Review project deliverables and assess completion	2 days	2024/12/19	2024/12/20	
- Conduct final testing and validation of AI features	1 day	2024/12/21	2024/12/21	27
- Evaluate project performance against initial goals and objectives	1 day	2024/12/22	2024/12/22	28
Documentation and Knowledge Transfer	3 days	2024/12/23	2024/12/25	29
- Document lessons learned and project outcomes	2 days	2024/12/23	2024/12/24	
- Transfer knowledge to maintenance team	1 day	2024/12/25	2024/12/25	30
Project Handover	3 days	2024/12/26	2024/12/28	31
- Deliver final project to the bank's digital banking team	2 days	2024/12/26	2024/12/27	
- Ensure smooth transition and provide post-implementation support	1 day	2024/12/28	2024/12/28	32
Project Closure Report	3 days	2024/12/29	2024/12/31	33
- Prepare project closure report	2 days	2024/12/29	2024/12/30	
- Obtain formal sign-off from stakeholders	1 day	2024/12/31	2024/12/31	34

Table 1 – Milestone Chart

4. Project Time Management

4.1. Introduction

Project Time Management refers to the process by which project management ensure that activities and tasks are well planned, scheduled, and controlled in the execution of deliverables. Effective time management will help in meeting the deadlines of project execution and assure maximum use of resources for timely enhancements to add value toward improving customer experience and satisfaction in Araliya Bank's implementation of its AI-powered online transaction platform.

It involve the development of a detailed schedule of the work into smaller activities, and then establishing the timelines for each of those activities. Monitoring is usually done with Gantt charts, Critical path method, and resource allocation plans to see the progress and delays.

Regular updating and proactive adjustments in the schedule are very crucial in order to keep it aligned with the objectives of the project.

Project Time management is an indispensable element that ensure the project will be successful since it reduces delays and overheads. It aids the project team to plan the work properly, thereby making the events fall between milestones and deploy a fully functional platform within the schedule time. This is significantly critical in the competitive banking space where timely deployment of innovative features considerably enhance customer satisfaction and loyalty.

By continuing to apply robust principles of time management, Araliya Bank is well-placed to ensure that its AI-powered platform will provide an effortless and time-saving, customer-oriented experience that upholds its reputation for digital banking services.

4.2. Task Breakdown and Sequence

Project Timeline -: 62 Days

- **Initiation Phase (10 Days)**
 - » **Project Charter Development (3 Days)**
 - Define project objectives (1 day)
 - Identify stakeholders and project team (1 day)
 - Establish high-level project scope and goals (1 day)

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- » **Feasibility Study (4 Days)**
 - Evaluate technical feasibility (2 days)
 - Assess financial feasibility (1 day)
 - Determine resource requirements (1 day)

- » **Initial Risk Assessment (3 Days)**
 - Identify potential risks (1 day)
 - Assess risk impact and likelihood (1 day)
 - Develop initial mitigation strategies (1 day)

- **Planning Phase (16 Days)**
 - » **Project Plan (7 Days)**
 - Determine resource requirements (1 day)
 - Create timeline with milestones and deadlines (2 day)
 - Develop budget and resource allocations plan (2 days)
 - Define project roles and responsibilities (1 day)
 - Establish communication plan (1 day)

 - » **Research Work (6 Days)**
 - Conduct market research on AI banking platforms (2 days)
 - Study, customer preferences and requirements (1 day)
 - Review legal and regulatory requirements (2 days)
 - Investigate technology option for AI integration (1 day)

 - » **Process Completion (3 Days)**
 - Finalize scope, schedule, and resources (1 day)
 - Obtain stakeholder approval (1 day)
 - Document the finalized project plan (1 day)

- **Execution Phase (23 Days)**
 - » **Data Collection (10 Days)**
 - Collect customer feedback and usage data from current platform (3 days)
 - Gather technical requirements for AI integration (4 days)
 - Compile security and compliance data (3 days)
 - » **Data Analysis (13 Days)**
 - Analyze customer behavior and preferences (3 days)
 - Analyze transaction data for patterns and optimization opportunities (3 days)
 - Evaluate AI algorithms effectiveness based on collected data (4 days)
 - Prepare insights and recommendations for platform improvements (3 days)
- **Closure Phase (13 Days)**
 - » **Final Review and Evaluation (4 Days)**
 - Review project deliverables and access completion (2 days)
 - Conduct final testing and validation of AI features (1 day)
 - Evaluate project performance against initial goals and objectives (1 day)
 - » **Documentation and Knowledge Transfer (3 Days)**
 - Document lessons learned and project maintenance outcomes (2 days)
 - Transfer knowledge to maintenance team (1 day)
 - » **Project Handover (3 Days)**
 - Deliver final project to the bank's digital banking team (2 days)
 - Ensure smooth transition and provide post implementation support (1 day)
 - » **Project Closure Report (3 Days)**
 - Prepare project closure report (2 days)
 - Obtain formal sign-off from stakeholders (1 day)

4.3. Time Management Plan (Gantt Chart)

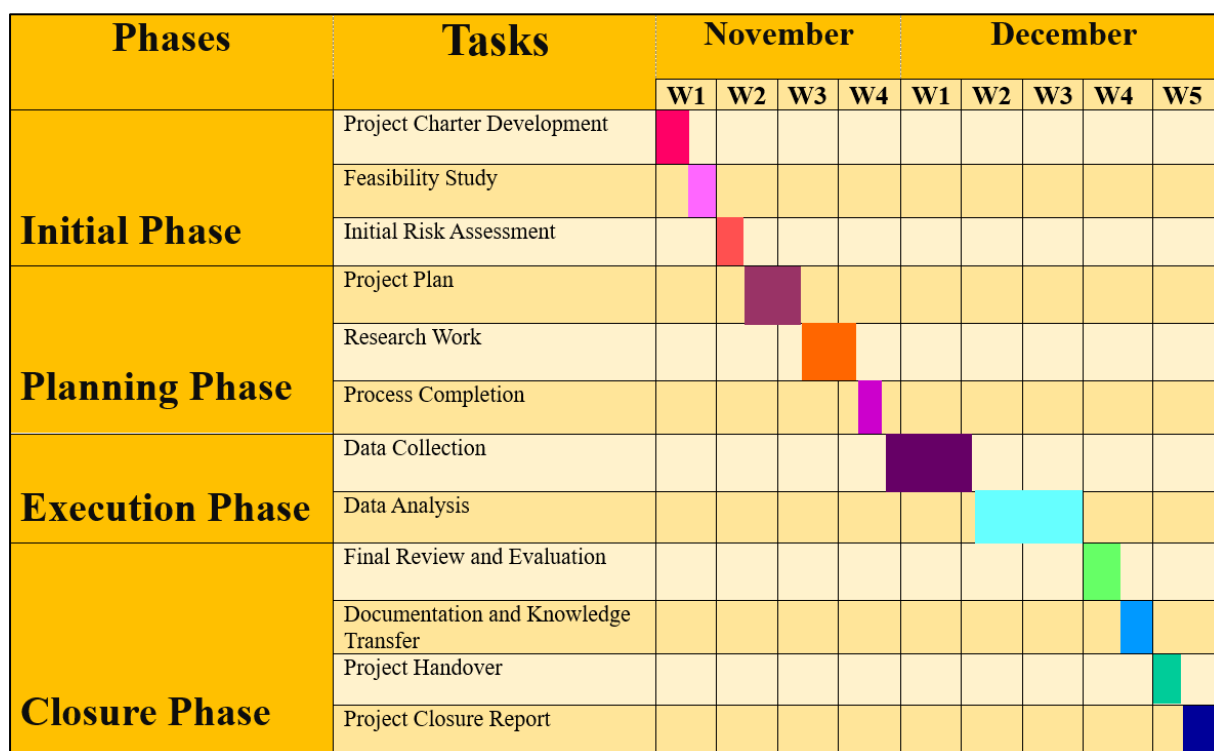


Figure 5 – Gantt Chart

5. Project Cost Management

5.1. Introduction

Cost management in a project is the most important component of project management, including all cost planning, estimating, budgeting, and controlling with a view to achieve completion of a project within an approved budget. In that regard, Araliya Bank's AI-powered online transaction platform needs effective cost management to each the desired project outcomes while maintaining financial viability.

The cost management planning of a project should start with the development of a cost management plan that describes how project costs precise cost estimation, based on reliable data and through analysis, helps in determining the financial resources required for implementing involves the assignment of costs to various components of the project to allow resources allocation aligned with project goals.

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Cost control is also equally important, it helps in ensuring that there will not be any budgeted costs and implementing corrective actions when necessary. In this project, infrastructure, and personnel are optimized to achieve the highest return on investment.

Effective cost management in projects will help Araliya Bank avoid unnecessary financial risk, ensure the platform is launched within the budget, and increase the impact of the AI-powered solution on customer experience. This also helps in ensuring that the project meets its financial targets and contributes toward overall success in enhancing digital banking services. Proper cost management not only supports financial stability but also fosters trust and confidence among stakeholders.

5.2. Cost Estimation

Category	Cost Items	Description	Estimated Costs (LKR)
Personnel	Project Manger	Pay for the duration of the project	80, 000
	Technical Team	Team member salaries (AI, ML, DA, etc.)	87, 000
	Research Team	Research and data collector salaries	75, 000
	Administrative Team	Administrative assistance salary	66, 000
Software and Equipment	Software Licenses	AI/ML software, AI tools	60, 000
	Hardware	Computers, Servers, and Storage devices	62, 000
	Cybersecurity Tools	Software for data protection	57, 000
Survey Tools	Online survey Platforms	Survey tools subscription fees	12, 000
Data Collection	Internet	Secondary data purchase	20, 000
	Travelling	Stakeholder meetings and data collection travel costs	30, 000

Training and Development	Seminars and workshops	Team training on new technology	20, 000
Operation Costs	Supplies for the office	Standard office supplies	12, 000
	Communication	Phones, Internet, and other communication tools	25, 000
Reporting and Documentation	Production of report	Report printing and binding	10, 000
	Presentation resources	Making resources for presentations	15, 000
Emergency	Emergency fun	10% of the overall budget is set aside for extra expenses	60, 000
Total Estimated Budget			691, 000

Table 2 – Cost Estimation

The entire outline of this budget is LKR 691, 000 and its distributed into the most vital cost categories, personnel, equipment, data collection, training, and contingency. The costs of personnel encompass the required remunerations to be given for the project manager, the technical team, the research, and the administrative support all adequate compensation for the man force. This includes all equipment and software licenses with tools of cybersecurity. The plan also make room for emergency expenses, furth strengthening the project against uncertainties. The bank intends to use thus budget effectively to successfully put into practice an AI online transaction platform, improving the customer experience.

6. Project Quality Management

6.1. Introduction

Project Quality Management is the process of ensuring that a project meet the desired standards to satisfy customers. It involves planning, assurance, and control processes to ensure that all aspects of the project are of the highest quality possible. In relation to the Araliya Bank Digital Banking project, Project Qualia Management is important delivering a seamless, secure, and efficient AI-powered online transaction platform. The project, through active quality management practices, ensure the working of the platform for its customers, meet their expectations, and also meets the required regulatory standards. This further leads to successful outcomes of the project through the achievement of stakeholders' expectations by providing a useful and reliable service.

6.2. Definition of quality standards and criteria

Quality Standards -: These are documented requirements, guidelines, or specifications that define the expected quality standards for any product or service. The standards ensure the output is within the defined expectations in performance and safety measures. In regard to Araliya Bank's digital banking platform, the quality standards could include secure data transmission, ease of use, and adherence to financial regulations.

The actual measures or attributes defining the quality standards realized comprise quality criteria. It might include system uptime, response times, user satisfaction, security of transactions, and the ability of the platform to handle different customer needs. Having clear

quality criteria will enable the project team to objectively measure the quality and success of the platform.

6.3. Implementation of Quality Assurance and Quality Control Measures

Quality Assurance concern the nonoccurrence of defects and effectiveness of processes applied to manage and create the product. In other word, regarding Araliya Bank's digital platform, the measures for ensuring quality may include.

- **Process Documentation** -: Well-definr processes through which the platform is being developed, tested, and maintained to attain consistency and efficiency .
- **Standard operating procedures** -: Detailing standardized operating procedures that should be followed by developers, engineers, and customer support to achieve uniformity.
- **Training and Skill Development** -: A well-trained staff in current technology, security protocols, and current best practices for customer service.
- **Regular Audits and Reviews** -: Running audits and processes from time to time to review area that need improvement, followed by adherence to quality standards.

Quality Control mainly delas with the detection of defects in the finished product to ensure that the platform meets the defined quality standards. Quality Control measures for Araliya Bank's platform could include,

- **Testing and Bug Fixing** -: Periodic testing of the platform, such as unit testing, integration testing, and user acceptance testing, to detect any functional, security, or performance issues in it. This includes using AI tools to detect bugs in real-time.
- **Monitoring Performance** -: Continuous monitoring of speed, uptime, and load performance, ensuring the platform is performing as per expectations.

- Feedback -: Taking feedback from customers to realize changes in usability that might improve overall customer satisfaction.
- Security Vulnerability Assessment -: Periodic scanning and penetration for vulnerabilities to eliminate all chances of security risks affecting customer data and trust.

Together, these QA and QC measures will make the Araliya Bank platform reliable and secure to use, thus meeting the expectations of customers for its long-term success.

7. Risk Management

7.1. Introduction

Project Risk Management involves the identification, analysis, and response to any risk that might pose a threat to the successful completion of the project. In this context, it is very important for the successful completion of the AI-powered online transaction platform of Araliya Bank by proactively overcoming various challenges that may either disrupt operations or hurt customer satisfaction. Risk management would ensure that cybersecurity, technology failures, or issues of user adoption are at their minimal or mitigated. This approach will help in maintaining the platform's reliability, improving customer trust in using the bank's services, and achieving the bank's goal of improving customer experience through innovative digital solutions. In this way, the project ensures it is sustainable and will have long-term success.

7.2. Risk Identification

We can separate two things to risk identification.

- Qualitative Risk
- Quantitative Risk

7.2.1. Qualitative Risk

Qualitative Risk identification refers to a non-quantitative analysis of risks based on possibility and potential damage. Some qualitative risk for Araliya Bank's AI network online transaction platform are as follows,

- Customer resistance to technology -: The customer would not be operate or even use the AI-based systems due to a lack of technical knowledge or even trust in digital platforms.
- Operations Risks -: Data breaches or system errors may affect the bank's reputation negatively.
- Regulatory non-compliance -: Inability to adhere to financial regulations may lead to penalties and dissatisfaction among customers.

7.2.2. Quantitative Risks

Quantitative Risk identification identifies risks numerically with data, to understand how the risk might affect the objectives of the project. Some quantitative risk for Araliya Bank's AI network online transaction platform are as follows,

- System Downtime-related Financial Loss -: The estimation of monetary effects due to the unavailability of services for every single hour.
- Fraud Incidents -: Estimating the amount that could have been lost owing to fraud cases stopped or not detected by AI.
- Customer Dropout Rate -: Calculating the number of customers lost based on system faults or dissatisfaction.
- Costs of Implementation -: Assess the risk of cost overruns while deploying the platform owing to unexpected expenses.

7.3. Risk Monitoring & Control Plan

A risk monitoring and control plan needs to be put in place for identifying, tracking and mitigating risks at all levels in the life cycle of Araliya Bank's AI-powered online transaction platform to proactively manage risks and reduce their impact on customer experience and satisfaction.

7.3.1. Important element of the Plan

- **Risk Tracking**

- » Record all identified risks in a risk register with the probability, impact, and mitigation strategies.
- » Update it periodically to show the present status of the risk

- **Performance Metrics**

Identify and institute key risk indicators, including,

- » System uptime percentage
- » Customer complaint rates
- » Fraud detection success rates

- **Regular Risk Review**

- » Hold periodic risk review meetings with stakeholders to take stock of progress in mitigating risks and emergent threats.
- » Utilize appropriate tools, such as dashboards and reports, for effective visualization of risk trends.

- **Mitigation Action Plan**

Develop action plans for each identified risk, including,

- » Delineate an escalation process when the risk involves high priority
- » Create backup and disaster recovery processes to reduce downtime to a minimum

- **Role and responsibilities**

Clearly outline specific roles regarding risk monitoring and control,

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- » Risk manager -: This person should oversee the whole process.
- » IT team -: Monitor system related risks and implement technical fixes.
- » Compliance officer to ensure that regulatory standards are met

7.3.2. Monitoring Tools and Techniques

- AI-driven analytics for real-time tracking of risk
- Use scenarios analysis and simulation to test the effectiveness of risk responses.
- Periodically audit the risk management process for further improvement

This plan ensures that through regular monitoring and control, risk that could be present are mitigated in time to enable smooth deployment and operational with the highest customer satisfaction level.

8. Communication Management

8.1. Introduction

Project communication management is the critical process of information exchange among stakeholders of the project at different instances of the project at different instances of its life cycle. In this regard, the AI-powered online transaction platform for Araliya Bank will be hugely dependent on communication management to align the goals of the development team, stakeholders, and end-users. Communication management to align the goals of the development team, stakeholders, and end-users. Communication management ensures that the project holds transparency, misunderstanding are lessened, and decisions are taken in a timely manner for the overall success of the project.

Clear and effective communication helps to address customer feedback, resolve technical challenges, and maintain stakeholder trust. Project communication management is important in fostering collaboration and keeping all parties informed to deliver a platform that will meet customer expectations, enhance user satisfaction, and driven the objectives of the project to completion.

8.2. Communication Management Plan

The communication Management plan spells out how the information will be communicated in terms of strategies, methods, and tools throughout the development and implementation of Araliya Bank's AI-driven online transaction platform. The plan ensures all stakeholders are well-informed, aligned, and engaged throughout the life cycle of a project.

❖ Objectives

- » Facilitate effective, consistent, and timely communication among all stakeholders
- » Ensure that project goals, progress, and changes are clearly communicated
- » Respond effectively to customer feedback and integrate it into the project

❖ Stakeholders

The importance stakeholders are,

- » Project Manager
- » Development Team
- » Senior Management
- » Customer Support Team
- » End-users

❖ Communication Methods

- » Meetings -: Weekly team meetings and monthly stakeholder meetings.
- » Reports -: Progress reports, performance updates, and issues tracking
- » Digital Tools -: Email, project management software (Microsoft teams and dashboards)
- » Customer Feedback channels -: Surveys, reviews, and direct customer support interactions.

❖ Communication Frequency

- » Daily -: Development team stand-ups for task updates.
- » Weekly -: Progress updates to project managers and key team leads.
- » Monthly -: Detailed reports for senior management and stakeholders

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Business Case

1. Introduction to Araliya Bank

Araliya Bank PLC was incorporated as a Public Limited Liability Company on 10th March 1986 and commenced its operations on 2nd April 1986. It is listed on the Colombo Stock Exchange. Since its inception, Araliya Bank has been in the forefront of technological innovations in the banking industry in Sri Lanka. Specifically, in 1988, it introduced Sri Lanka's first multi-point network of automated teller machines (ATMs), popularly known as "Araliya Electronic Teller" or "SET." The bank continued its pioneering efforts by introducing South Asia's first debit card in 1997 and establishing Sri Lanka's first Cheque Imaging & Truncating (CIT) site in 2004.

Currently, Araliya Bank is operating a total of 229 branches and 373 ATMs within the country to provide a complete range of banking services throughout the country in 2023. Business operation of the bank mainly comprises retail and commercial banking. The bank has always tried to modernize itself with technology in ways that serve customers better. It has been able to introduce a number of services such as Internet Banking, Mobile Banking, and Slip Less Banking.

In 2022, Araliya Bank reported revenues of LKR 206,002 million and net income of LKR 14,061 million. Additionally, the bank had assets of LKR 1.367 trillion and equity of LKR 135.551 billion. All these indicate a very important position in the structure of Sri Lanka's financial system.

Araliya Bank has been accredited as a leader in the banking sphere, with several awards. It was recognized as the "Bank of the Year" by The Banker of Financial Times Limited, London, for two consecutive years, and it has been bestowed with the "National Business Excellence Awards 2010."

In the fiercely competitive Sri Lankan banking industry, Araliya Bank has positioned itself as one of the strong players-it was the third-largest private sector bank in terms of total assets in 2009. Its commitment toward innovation and customer-centric services propels it further in growth and market presence. (internationalbanker, 2016)

1.1. Araliya Bank Organization Structure

The organizational structure of Araliya Bank is built to provide effective management and smooth operation,

❖ **Board of Directors**

The Board, which is made up of the Chairman and a number of independent and non-executive directors, is in charge of the bank's financial performance, compliance, and strategic direction.

❖ **Executive Management**

- » **Chief Executive Officer (CEO) :-** In charge of carrying out the Board's plans and managing the bank's general operations.
- » **Deputy General Manager (DGMs) :-** Oversee key operational divisions, including risk management, IT and Digital banking, Retail Banking, and corporate banking

❖ **Senior Management**

Consists of Assistant General Manager (AGMs) who are in charge of particular divisions such Branch Operations, Marketing, Compliance, and Treasury.

❖ **Middle Management**

At the branch and divisional levels, branch managers and department heads make ensuring that strategies are carried out effectively.

❖ **Operational Staff**

Tellers, relationship manager, IT staff, and other frontline employees and support staff provide daily services to clients.

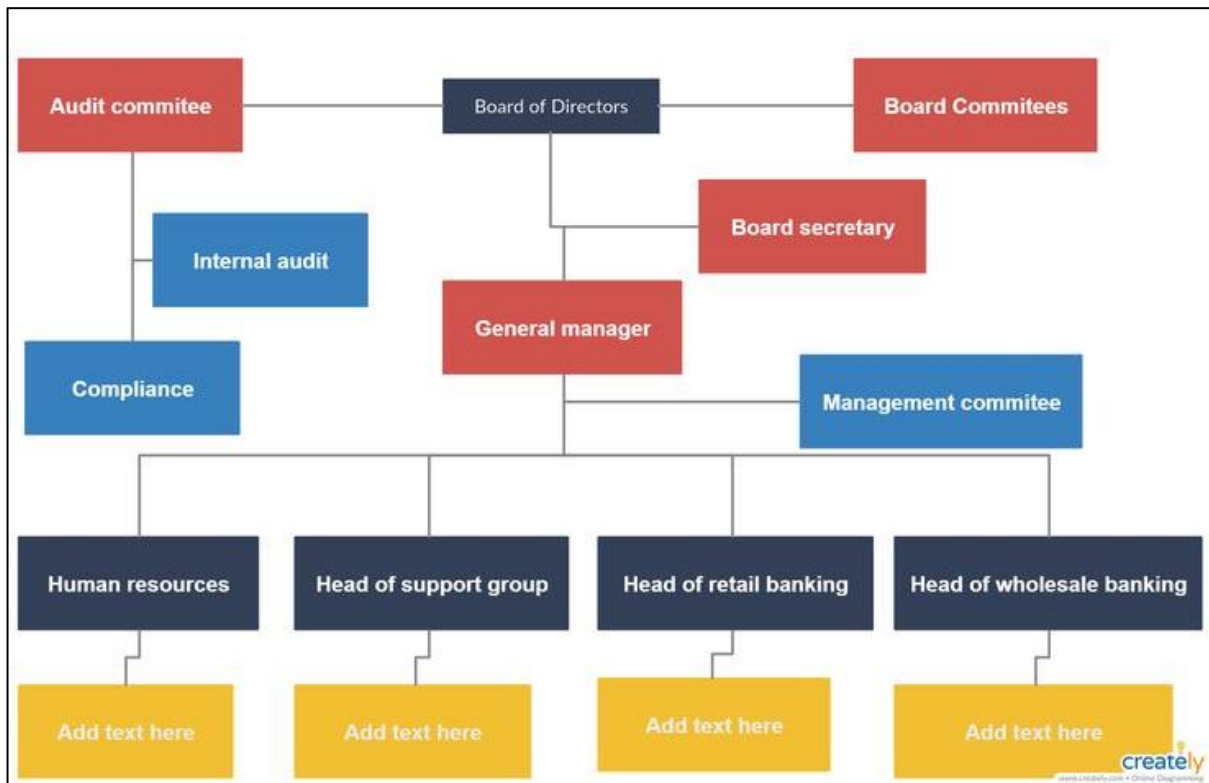


Figure 6 – Organization Structure

(Contributioms, 2024)

1.2. Core Business Operations of the Araliya Bank

Business-wise, Araliya Bank is primarily involved in various banking and other financial services for both retail and corporate clints. Essentially, it reinvented itself for the needs of a technologically enabled market while continuing access to basic banking services for a wide range of customers.

1.2.1. Overview of the organization's primary products or services

Araliya Bank offers the below primary products and services,

❖ Retail Banking

- Savings and Current Accounts -: Comprises of personal savings, current accounts, or any variants therefore at market process with electronic access to a broad array of online services offered.

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- **Loan and Mortgages -:** This involves personal loans, housing loans, and car loans designed in a way to assist him in meeting his monetary requirements.
- **Cards -:** It involves various types of credit card options aimed at pleasing the customer through spending habits, desired reward programs, rebates, and flexibility.
- **Digital Banking Services -:** This involves internet banking, mobile banking, and e-wallets through which the customers can carry out transactions, pay bills, and manage financial matters literally anywhere in the world.

❖ **Corporate Banking**

- **Business Accounts -:** From accounts and services relating to SMEs to large corporations
- **Loans and Trade Finance -:** Finances for business expansion in the form of working capital loans, trade finance, and project financing.
- **Merchant Services -:** Enabling the payment through merchant services and point of sale systems to make each transaction effective and safe.
- **Treasury and Investment Services -:** The bank offers foreign exchange, structured finance, and investment solutions to corporate clients.

❖ **Investment Banking**

- **Corporate Advisory Services -:** Providing advice on merger, acquisition, and corporate restructuring services.
- **Capital Markets -:** It refers to the services of equity and debt capital markets that help business raise funds by issuing stocks among other instruments

❖ **Wealth Management and Insurance**

- **Wealth Management Services** -: The bank offers tailored solutions to help individuals manage their investments, retirement funds, and estate planning.
- **Insurance products** -: Araliya Bank is in collaboration with insurance providers to offer a range of personal and corporate insurance products.

1.2.2. Description of the target market or customer base

- **Individual Customers** -: It offers the gamut of products beginning with a basic savings account to more sophisticated wealth management services catering to personal financial needs.
- **Small and Medium Enterprises (SMEs)** -: It offers financial products for businesses in the form of business loans, credit facilities, and cash management solutions.
- **Corporates and Large Businesses** -: Large businesses benefit from corporate banking products including trade finance, working capital financing, and investment banking.
- **High-net-worth individuals (HNWIs)** -: The bank's wealth management services target affluent individuals seeking customized financial and investment solutions.

1.2.3. Araliya Bank's value proposition or unique selling points

- **Technologies Innovation** -: The bank pioneered digital banking solutions in Sri Lanka, including mobile and internet banking platforms, in order to provide increased convenience and accessibility to its customers.
- **Customer Centric Services** -: Araliya bank offers personalized banking to both retail and corporate customers through dedicated relationships managers who ensure that the customers get tailor made financial solutions.

- **Security and Trust -:** The bank has been known for its high-security levels in transactions, supported by advanced encryption technologies and fraud detection systems.
- **Local expertise and market knowledge -:** With deep understanding of the Sri Lankan market, Araliya Bank provides solutions that are relevant and suited to the local economy

1.2.4. Key partnerships or collaborations that support the core operations

- **Telecommunication Partnerships -:** Araliya Bank has tied up with telecommunication companies to offer mobile banking and e-wallet services, enabling a seamless transaction facility through mobile phones.
- **Insurance Partnership -:** The bank has partnered with leading insurance companies to offer its customers a suite of insurance products, from life health insurance to corporate cover.
- **Internal Payment Network -:** In order to offer international banking services, including credit and debit card payments, Araliya Bank has partnered with international payment networks such as Visa, MasterCard, and UnionPay.
- **Global Banking Relationships -:** The bank maintains relationships with correspondent international banks and financial institutions in facilitating foreign exchange, cross border transactions, and international trade finance.

1.3. Organizational Data

1.3.1. Structured, Semi Structured and Un-Structured data generated by Araliya Bank

Araliya Bank generates and processes various types of data to support banking operations, customer services, and business decisions. Based on the classification of the structure and organization of these data types, it may fall under structured, semi-structured, or unstructured data.

➤ **Structured Data**

Structured data is highly organized and hence easily stored in relational databases or spreadsheets. It generally follows a predefined format with rows and columns. E.g.,

- Customer Information -: The name, address, account number, and transaction history of customers are stored in tables within the bank's CRM systems.
- Transaction Data -: These include the record of all the financial transactions deposits, withdrawals, transfer, and payments all in a structured manner with detail in amount, account number, date, and type.
- Loan and mortgage data -: Structured records of loan amounts, interest rates, repayments schedules, and customer profiles.
- Employee Records -: This covers details such as names, position, salary, and performance ratings of employees.

➤ **Semi-Structured Data**

Semi-structured data does not have rigid structured that structured data has, but it does contain a few tags or markers that separate the elements and might make it easier to process. Most of the time, semi-structured data will store data in formats such as XML, JSON or NoSQL database. E.g.,

- Email and correspondence -: The metadata and actual content are usually stored in semi-structured database formats like JSON or XML.

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- Transaction Logs -: While transaction logs can be semi-structured, they do contain valuable information that might follow a semi-organized format (e.g., system logs that record user interactions, login activities, and payment verification events)
- Customer Feedback -: The feedback forms or customer surveys may be stored in semi-structured format where customer sentiments are captured with tagged responses in different fields.

➤ **Unstructured Data**

Unstructured data is basically any form of data that does not have a defined format or organization it's thus very difficult to store and analyze. More advanced tools and technologies are needed to process this kind of data. E.g.,

- Social media data -: Comments , posts, and engagements on Facebook, Twitter, or LinkedIn about Araliya Bank are all examples of unstructured data that give insight into customer satisfaction, brand sentiment, and market trends.
- Text Documents -: Most internal reports, customer complaints records, contracts, and documents of term and conditions are usually unstructured, so special document management systems or NLP tools are needed to draw useful information.
- Multimedia Data -: Recorded calls with customers, training videos, or marketing materials in the form of audio, video or image files are, by nature unstructured and kept on a file system or in cloud storage.
- Chatbot Interactions -: Conversations with AI-controlled chatbots or customer support system create unstructured data that can be analyzed in order to find out customer queries, sentiment, and feedback.

1.3.2. How Araliya Bank Stores Structured, Semi-structured, and Unstructured Data

➤ Storage of Structured Data

- Relational database -: Araliya Bank uses an enterprise level relational database like Oracle database, Microsoft SQL Server to store structured data, which includes customer records, details of all transactions, and employee details. In these databases, data is organized in tables with predefined schemas for easy querying and reporting.
- Data Warehouses -: Araliya Bank can make use of data warehouses such as Amazon Redshift or Teradata to store huge amounts of structured data for analysis and decision-making, which can be queried and reported efficiently.

➤ Storage of Semi-Structured Data

- NoSQL -: Technologies like MongoDB, Cassandra, and Couchbase provide a better scenario for storing semi-structured data, which, very much like transaction logs, customer interactions, and metadata from various sources (e.g., emails or chat logs).
- Documentation Management Systems -: Araliya Bank will use SharePoint or Google Drive system for storing semi-structured documents with metadata tagging capabilities, which will enable easy retrieval and organization of file such as contracts, invoices, and correspondence.

➤ Storage of Unstructured Data

- Object Storage -: The scalable cloud based storage, such as Amazon S3, keep unstructured data information from applications dealing with or using audio, video recordings, images, and documents.

- **File systems -:** More traditional types of file systems, on premises or cloud based, can store large collections of unstructured data, such as customer call recordings or training materials organized by folder structure and file naming.
- **Big Data Technologies -:** More sophisticated processing can be performed on unstructured data using big data technologies such as Hadoop or Apache Spark, which enable the storage and processing of diverse unstructured datasets on a large scale.

1.4. Systems used in the Araliya Bank

Araliya Bank employs a number of systems that help automate its operations, enhance customer experiences, and ensure effective management of resource. These systems are vital in ensuring day to day smooth operations, meeting all the regulatory requirements, and innovating in the financial services sector. Below are some of the key systems used in the organization.

System	Description	Features	Example
Core Banking System (CBS)	A centralised system for handling deposits, loans, transactions, and accounts.	Processing transactions in real time, managing accounts, and adhering to regulations.	Finacle, Temenos T24
Enterprise Resource Planning (ERP)	Including business operations such as compliance, procurement, and finance	Procurement, financial management, and reporting	Oracle ERP Cloud, SAP S/4HANA

Human Resource Information System (HRIS)	Oversees HR operations, including hiring and performance reviews, and employee data	Payroll integration, attendance monitoring, and employee records.	Oracle HCM, Workday
Payroll System	Guarantees correct wage processing and adherence to labour regulations	Payslip creation, tax compliance, and salary computation.	Oracle Payroll
Customer Relationship Management (CRM)	Oversees consumer preferences, relationships, and offer individualised services.	Marketing automation, service tracking, and customer segmentation.	Microsoft MetricStream, IBM OpenPages
Enterprise Risk Management (ERM)	Identifies, evaluates, and reduce operational risks.	Risk evaluation, compliance tracking, and reporting	MetricStream, IBM OpenPages
Business Intelligence (BI) and Analytics	Gathers information for performance insights and decision-making	Predictive analytics, trend analysis, and data visualisation.	Tableau, Power BI
Document Management System (DMS)	Maintains compliance when storing and retrieving documents	Retrieval, access control, and document storage	DocuSign, Microsoft SharePoint
Mobile Banking System	Offers banking services using mobile devices	Real-time alert, account management, and mobile payments	Flutter, Built using React Native

Online Banking System	Permits online financial transactions	Account administration, money transfers, and bill payment	Fiserv, Temenos Infinity
Security Systems	Safeguards private client and company information	Access control, fraud detection, and encryption	Symantec, Checkpoint, McAfee
ATM and POS systems	Enable payments, fund transfers, and cash withdrawals through POS and ATM terminals	Fund transfers, bill payments, and cash withdrawals.	Diebold Nixdorf, NCR

Table 3 – Systems used in Araliya Bank

1.5. Identification of Stakeholders

Stakeholders can be defined as individuals, group, or organizational entities having a stake in or being affected by the operation and performance of Araliya Bank. They are very important to the bank, as they determine the way forward, policy, and growth of the bank. Stakeholders could generally be classified into two categories,

- Internal
- External

1.5.1. Internal Stakeholders

Stakeholders Group	Role and Interest
Employee	• Lead every operation, innovation, and customer service
	• Seek employment stability, career advancement, and just salary
	• Contribute to the productivity and client happiness of the bank

Managers and Executives	• Create and implement business plans
Shareholders	• Assure operational effectiveness and adherence to rules.
	• Aim to increase profits and accomplish organisational goals
	• Provide fund to support the bank's expansion and operations
	• Anticipate long-term sustainability, higher share value, and financial benefits.

Table 4 – Internal Stakeholders

1.5.2. External Stakeholders

Stakeholder Group	Role and Interest
Customer	• Make use of the bank's offerings, including loans, savings accounts, and outline banking
	• Anticipate dependable, safe, and easy-to-use financial solutions
	• Aim for outstanding customer service and confidence when doing financial transactions
Suppliers	• Supply infrastructure, office supplies, and technology, among other necessities
	• Anticipate long-term collaborations and prompt payments
Government Agencies	• Enforce legal requirements, tax observance, and banking rules.
	• Anticipate that the bank will support economic expansion and abide by both domestic and international rules.

Communities	• Represent the environmental and social context in which the bank works
	• Anticipate corporate social responsibility (CSR) projects including community welfare, sustainability, and education
Investors	• Provide the bank more money and resources
	• Anticipate monetary stability, openness, and lucrative rewards on investment
Competitors	• The market's other banks and financial organisations
	• Encourage innovation in the sector and have an impact on market dynamics

Table 5 – External Stakeholders

For Araliya Bank, it is essential to comprehend the requirements and expectations of both internal and external stakeholders. The bank can keep its competitive advantage, improve its reputation, and achieve sustainable development by attending to customer issues and cultivating beneficial connections.

1.6. Impact of Stakeholders on the Araliya Bank Success

The stakeholders of Araliya Bank play a very instrumental role in creating its success. Each stakeholder group has some kind of influence on the organization through their, expectations, and involvement. Below is a stakeholder matrix and explanation of the impact of internal and external stakeholders.

1.6.1. Stakeholder Matrix

High Power/High Interest	High Power/Low Interest
Executive/Managers	Stakeholders
Regulators/Government Agencies	Investors
Key Customers	
Low power/High Interest	Low Power/Low Interest
Employees	Local Communities
Suppliers	Competitors

Figure 7 – Stakeholders Matrix

1.6.2. Important Internal Stakeholders and their Effects

- **Employee**
 - » Promote operational effectiveness and guarantee top-notch client support
 - » Increased creativity and productivity are a direct result of improved morale and skill development.
- **Managers/Executives**
 - » Create and implement plans in line with the objectives of the company.
 - » Assure adherence to rules and efficiency control risk
- **Shareholders**
 - » Provide the funds required for the expansion and growth of the organization
 - » Their trust in the bank's financial success has a direct effect on its reputation and market value.

1.6.3. Important External Stakeholders and their Effects

- **Customers**
 - » Their use of banking goods and services has a direct influence on revenue
 - » Retention, loyalty, and word-of-mouth recommendations are all influenced by positive client experience.

- **Government Agencies**
 - » Regulatory compliance guarantees that the bank stays out of trouble and does business lawfully
 - » Have an impact on the bank's policies, especially those pertaining to digital banking, taxation, and risk management.
- **Suppliers**
 - » Provide the necessary infrastructure, such as security and IT systems.
 - » Customer satisfaction and operational stability are improved by effective supply networks.
- **Investigators**
 - » More money should be allocated for major expansions or projects
 - » Good returns on investment help to trust in the bank's strategy and leadership
- **Communities**
 - » Anticipate CSR initiatives, completing them enhances the bank's reputation and fosters trust

1.7. Analyze Challenges to the Success of a Business

- **Market Competition**

Discussion -: The banking industry faces stiff competition from established financial institutions, fintech companies, and alternative digital banking platforms. Araliya Bank has to offer its products and services with differentiated features such as AI-powered services, competitive pricing, and superior customer experience.

Impact -: High competition driven innovation but also pressures margins. A failure to maintain market share will decrease both customer base and revenue.

- **Economic Factors**

Discussion -: Inflation, changing interest rates, and exchange rate fluctuations directly impact how customers spend and borrow. Further, macroeconomic trend may also affect operating costs and profitability.

Impact -: In an adverse economic environment, revenues may get hurt as consumers cut back on discretionary spending. Increasing costs from inflation or exchange rates volatility can also further erode profitability.

- **Technical Advancements**

Discussion -: With the rapid changes in technology, Araliya Bank needs to continuously invest in upgrading its systems to match changed customer preferences for digital banking. Keeping pace with technological innovations such as blockchain, mobile-first technologies, among others, is very critical.

Impact -: While these investments enhance service delivery, they increase operational costs enormously. Filling to keep up with technological adoption means losing tech savvy customers to competitors.

- **Operational Inefficiencies**

Discussion -: Operational challenges in the form of supply chain disruptions, inefficient inventory management in the distribution of cash, or production bottlenecks in the launch of new services could hamper productivity.

Impact -: This will inefficiencies, late delivery, increased costs, and lower customer satisfaction, possible affecting loyalty and retention rates.

- **Human Resources Management**

Discussion -: The capability to attract, develop, and retain competent staff is central to quality service delivery. Diversity in the workforce and maintaining employee satisfaction in the face of changing expectations of work add to the complexity.

Impact -: High turnover rates or inadequate staff experience can result in inconsistencies in service and increased cost of recruitment, weakening overall customer trust.

- **Customer Retention**

Discussion -: Changes in customer preferences and rising expectations make loyalty difficult to maintain. Now customers expect a seamless digital experience, personalized services, and proactivity.

Impact -: Churn from unmet expectations can hit the top line. Competitive pressures demand constant innovation just to keep up with customer expectations.

- **Brand Reputation**

Discussion -: Maintaining a brand's reputation is essential, especially when dealing with ethical issues, consumer complaints, and dispute resolution. Service errors or cybersecurity breaches may result in bad press.

Impact -: A tarnished image may result in a drop in market value, lost revenue, and mistrust from customers. It is crucial to manage ethical behavior and public relations proactively.

- **Globalization and Trade Barriers**

Discussion -: The bank faces trade laws, geopolitical unpredictability's and difficulties when it enters foreign markets.

Impact -: Globalization raises operational complexity even as it presents chances for growth. Plans for expansion may be derailed and political unrest, which might have an impact on profitability.

- **Innovation and Adaptation**

Discussion -: A culture of constant innovation is required due to the banking industry's dynamic nature. Products and services may become outdated due to slow uptake or resistance to change.

Impact -: Competitiveness is weakened by a lack of innovation. On the other hand, encouraging innovation may establish Araliya Bank as a market leader and propel sustained expansion.

- **Risk Management**

Discussion -: Business continuity is seriously threatened by risks including natural catastrophes, cybersecurity attacks, and legal obligations. Frameworks for effective risk management are essential.

Impact -: Serious financial losses, fines from the government, and harm to one's reputation can result from inadequate risk mitigation techniques. Resilience against unanticipated interruptions is ensured by proactive risk assessment.

1. CHAPTER ONE – INTRODUCTION

1.1. Introduction

Artificial Intelligence (AI) now transforms industries of the world through process automation, enhancing decision-making, and operational efficiency. AI is the replication of human intelligence in machines, which enables them to perform tasks like learning problem-solving, and data analysis. This transformative technology has been taken up widely across sectors in health, manufacturing, and finance, where this is remodeling traditional business models and is creating innovative opportunities. Machine learning, natural language processing, and predictive analytics are considered the key AI technologies that drive efficiencies, improve customer interaction, and deliver firms' competitive advantage. AI has now proved to be very important in the financial industry for improving customer service, smoothing operation workflows, and ensuring sound security. AI-based solutions, like chatbots, provide real-time support to customers through faster and quick query responses. Predictive analytics allows for better decision -making on credit risk assessment and fraud detection. This will enable banks to offer tailored services, smooth their operations, and gain their customers' trust with the assurance of secure and efficient operations.

This given study will dwell on Araliya Bank's online transformation platform powered by AI and its contribution to the engagement and satisfaction of customers. In this respect, with digital banking on a fast respect, with digital banking on a fast pace, AI-powered solutions have become vital in fulfilling changing customer needs and ensuring competitive performance within the financial sector. Challenges regarding implementation costs, data privacy and ethics also need consideration. Therefore, this study is focused on understanding how to effectively integrate AI into the digital banking operations of Araliya Bank, considering the benefits of AI in banking services.

(Rashid, A.B. and Kausik, A.K. , 2024)

1.2. Purpose of Research

This research, therefore, aims at critically assessing the performance of Araliya Bank's online transaction platform, integrating AI into ultimate customer experience and satisfaction in the digital banking industry. Moreover, there will be an increased need to understand how this innovation in the use of AI technologies meets the increasing customers' demands in the financial services for speed, convenience, personalization, and security. Some of the challenges that this research work will seek to overcome include data privacy, system reliability, and the digital literacy divide that may hamper wide acceptance of the platform. The research will identify specific benefits and limitations of the AI system and provide insights for refining the features of the platform and improving overall customer engagement. In the future, it will align Araliya Bank's digital banking services in line with customer expectations and help the bank in its superior and inclusive financial services.

Artificial intelligence has transformed the banking industry by automating processes, enhancing decision-making, and making customer service more personalized. In Araliya Bank's digital banking, AI is very important in areas such as improved customer support, where AI-driven chatbots and virtual assistants answer questions and solve problems 24/7, reducing wait times and boosting customer satisfaction. Moreover, AI allows for personalization of banking with the analysis of customer behavior and transaction history to provide tailored financial advice, product recommendations, and promotions that improve engagement and loyalty. AI is also instrumental in fraud detection by facilitating real-time monitoring of transactions and the identification of unusual patterns that enhance security and prevent fraudulent activities. AI also ensures better operational efficiency since these systems can automate transaction processing, account management, and loan approvals. These significantly reduce human error and operational costs, freeing staff to attend to higher levels of tasks. While these improvements bring on better performance, variety of different barriers still exist to full utilization, data privacy concerns, limitations in the interpretation of complex queries by customers, technological constraints, the digital divide in literacy, and biases in AI-systems. These challenges will have to be overcome if the full power of AI is to be exploited within the banking sector.

Notwithstanding improvements brought by AI, many obstacles impede the full use of AI to benefit customers' experiences and industry performance. Major challenges persist on data privacy concern, AI requires lots of sensitive information about customers. It has to be well governed and protected with appropriate measures so that it will inspire trust in it. AI's inability to completely understand complex or ambiguous customer support. Implementing and maintaining such AI systems is a huge technological and infrastructure investment, thus becoming quite difficult for smaller institutions or even states that may not have the means. In addition, the success of AI platforms is based on customer adoption and the ability to use digital technologies. The digital divide in literacy could therefore prevent a big fraction of customers from accessing the services. Also, if an AI model is trained using biased data, biases in that model can show up as discrimination against certain classes of customers, hurting their experience and satisfaction.

1.3. Significance of the Study

The value of this research to the banking industry and customers lies in its comprehensive discussion of AI-powered online transaction platforms, with special attention to how Araliya Bank has implemented AI. The results can be generalized to optimize the adoption of AI within the results can be generalized optimized the adoption of AI within the results can be generalized to optimize the adoption of AI within the banking sector toward better customer experiences. The present study has provided practical insights that may help stakeholders like bank managers, policy frames, and technology developers to understand the pros and limitations of AI in a real-world scenario. Finding can be used strategically by the managers of banks to implement AI into operation for better customer service and efficiency. Policymakers will be able to create regulations that allow for responsible use of AI without violating customer privacy. Technology developers can further fine-tune the AI algorithms for better personalization and fraud detection, also enhancing real-time application performance. Finally, the study provides actionable guidance on how to optimize the use of AI in banking while highlighting areas that need improvement.

As well as this research will present an important addition to the current academic discussion on the use of AI in the context of the financial sector and the improving of the customers'

experience. Since it centers on Araliya Bank, the study offers concrete evidence of AI's application in digital banking and helps fill the gap in literature, which mainly covers the systems of entire worlds. The research will also provide a benchmark for subsequent research work by coming up with a model on the right deployment of AI in the banking sector. Besides, it will provide information concerning AI drawbacks, including the problem of algorithmic bias and data protection issues, which will create a comprehensive view of AI's capabilities. Last but not, this study will contribute to the existing body of knowledge on customer-centered AI adoption where issues such as requirements and expectations of customers will be adopted to influence the development of better human-centric applications of the AI technology in banking.

1.4. Research Objectives

1.4.1. Main Objective

The main objective of this research is therefore to assess the effectiveness of, and drawbacks of Araliya Bank AI implemented online transaction platform for the improvement of customer value. The purpose of this research is to determine the role of AI element and functionalities in enhancing the service quality and client engagement as well as customer satisfaction and identify the corresponding strengths and weaknesses.

1.4.2. Sub Objectives

- To identify the main advantages of Araliya Bank's AI-powered online transaction platform for enhancing client satisfaction.
- To evaluate the AI-powered platform's drawbacks and difficulties.
- To assess the degree of client satisfaction attained by utilizing Araliya Bank's artificial intelligence-driven platform.
- To investigate how AI may improve client loyalty and engagement in online banking.
- To make suggestions for enhancing the AI-powered platform in order to enhance the user experience even further.

2. CHAPTER TWO – LITERATURE REVIEW

2.1. Introduction

This review will, therefore, from the basis on which it will be possible to present the practical use and limitations of AI at Araliya Bank, particularly how AI has been able to influence customer experience. This chapter discusses prior research, findings, and deliberations on the role that AI plays within the banking industry in regard to customer centricity and processes of transactions. It also seeks to assess the status of AI adoption in banking today, analyzes benefits, and describes challenges that restrain the full development of AI in the banking sector. This review synthesizes data from various academic platforms shape the future of digital banking, especially in Araliya Bank.

The review shall cover a few areas of concern. It starts with the overview AI, “what is it”, “the core concepts in it”, and “how big the role of AI in changing many sectors”, especially banking: cover fraud detection, chatbots, or automation in customer service, predictive analytics in financial planning, and personalized customer experiences. Then, the review has thrown light on the benefits of accruing from the adoption of AI: increasing customer satisfaction, smoothing operational efficiency, and saving on costs. Further, the paper enumerates the constraints and challenges faced by the sector data privacy issues, high implementation cost, resistance to technology, and technical glitches, The review concludes by identifying the empirical gaps in the literature, with call for further research on effective strategies to optimize AI in the banking context, with particular emphasis on the Sri Lankan banking environment.

This chapter, therefore, provides the premises for the study by offering a critical review of the literature on both the possibilities and challenges presented by AI to Araliya Bank’s digital banking ecosystem. It further situates the research within the purview of extant scholarly and practical discussions while isolating issues that call for innovation and enhancement I the AI-driven banking system.

(lahi, 2023)

(Anon., n.d.)

2.2. Overview of Artificial Intelligence (AI)

Artificial Intelligence, or AI in short, is a field within computer science that designs systems capable of tasks normally requiring human intelligence. These include learning and reasoning, problem-solving, perception, understanding natural language, and decision-making. AI, therefore, enable machines to perform like human cognitive functions and embark on activities which hitherto might have been beyond the scope of machines with a view to improving efficiency and effectiveness.

Core concepts of AI include:

- **Machine Learning**

Systems that can learn from experience and get better without explicit programming thanks to a subset of AI. ML algorithms examine data, spot trends and come to conclusions or predictions based on the data.

- **Natural Language Processing**

A branch of AI that focuses on how computers and human language interact, allowing systems to understand, interpret and react to voice or text input.

- **Computer Vision**

The ability of AI to interpret and process visual information, such as images or videos to perform tasks like facial recognition and object detection.

- **Robotic Process Automation**

Automation of repetitive and rule-based tasks using AI, reducing human intervention in routine operations.

- **Predictive Analytics**

The use of AI to analyze historical data and predict future trends, enabling organizations to make informed decisions

(GeeksforGeeks, 2017)

2.3. Practical Applications of AI in the banking sector

- Application areas of AI within the banking business have totally shifted the traditional works, using newer technologies such as machine learning, NLP, and robotics automation. There is a redefinition of every way in which banking services are available to customers since AI-powered interfaces and virtual agents enabled customer service beyond anything previously envisaged due to their operation 24/7 with faster reply to mechanisms and even personalized interaction if needed. (Brown et al., -2021)
- Machine learning models greatly enhanced fraud detection and prevention through the identification of suspicious patterns and anomalies in real time, therefore reducing financial losses and enhancing customer trust. (Smith et al., -2020)

In addition, AI-powered credit risk assessment tools analyze large datasets for accurate credit scoring and thus facilitate more informed and speedy lending decisions. (Doe et al., 2019)

- AI also ensures operational efficiency with RPA, automating basic tasks such as data entry, reconciliation, and compliance monitoring. These innovations reduce not only manual workload but also minimize errors to ensure smooth banking operations. Predictive analytics powered by AI further helps in personalized financial planning and target marketing campaigns that helps improve customer satisfaction and loyalty. (Johnson & LEE, - 2022)

(Anon., 2025) , (Anon., 2025), (Anon., 2023), (Keith Reinhardt , 2024), (Anon., 2021)

2.4. Benefit of AI Adoption in the Bank

- Efficiency Gains and Cost Savings

Artificial Intelligence makes automation possible for a range of routine tasks, from transaction processing to customer onboarding and fraud detection. This significantly cuts on the needs for human intervention. Using this advantage, banks can now process more transactions with limited resources. (Green et al., - 2021).

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Also show that through AI-powered robotics process automation, human errors and operational costs are minimized, the workflow is optimized, which enhances the speed of service delivery.

- Improved Accuracy and Risk Management

Machine learning algorithms in credit scoring , fraud detection, and risk assessment offer more accurate predictions and insights to the banks. An AI system can analyze huge volumes of data identify patterns and trends that are often not detected by human analysts.

(White & Thomas, - 2022). This, in turn, leads more informed decision-making and reduces risks on lending, transactions, and investment.

- Scalability

AI solutions give banks the scalability they need to manage expansion without having to make corresponding increases in infrastructure or personnel. AI-powered systems may be scaled to match demand as transaction or customer contact volumes increase, guaranteeing reliable service delivery (Brown et al., - 2021). Without being constrained by conventional models, banks may extend their services to new geographic areas or clientele groups because to this scalability.

- Enhanced Customer Satisfaction

AI helps improve customer satisfaction by providing personalized experiences, quicker response times, and greater availability of services. AI-driven chatbots and virtual assistants provide 24/7 customer support, answering queries and resolving issues in records time.

(Lee et al., - 2023).

In addition, AI-enabled recommendations systems analyze the behavior and preferences of individual customers, enabling banks to offer financial products and services that best meet each customer's needs (Johson & Lee, - 2022). These personalized experiences raise customers loyalty and satisfaction, as customers feel that their banking needs are being proactive.

(Macrina Lazo, 2023), (Omar H.Fares, 2022), (Hengjun Mei, 2024), (Serkan Demirel, 2024), (Tierno, 2024), (Anon., 2021)

2.5. Limitation and Challenges

The adoption of AI in the banking sector is associated with various benefits, but it also poses a number of limitations and challenges. Some are under technical and sector-specific challenges, while others are economic, resource-related, or ethical in nature. These are important limitations that need to be addressed for AI solutions to make a difference with responsible improvements in customer experiences and operational efficiencies.

- **Technical Limitations**

Some of the main technical challenges that hinder the adoption of AI relate to the preconditions of large, high-quality datasets that train the AI models. Poor or biased data results in poor predictions and, as such, poor decision-making within credit scoring and fraud detection among other areas. (Smith et al., 2021).

Moreover, biases in AI algorithms, whether for historical data patterns or lack of sufficient training data, might lead to unfair outcomes, including discrimination against some groups of customers. The absence of interpretability, or “explainability,” of complex AI models stands as another challenge because it cannot be easily comprehended and justified by banks and regulators for the decisions made by these systems (Jones & Patel, - 2022)

- **Sector-Specific Issues**

Transparency is a crucial issue in the banking industry. Financial institutions need to operate within strict regulatory frameworks, which demand transparency in decision-making. Much of the AI systems being used are black-box in nature; thus, maintaining the required level of transparency, especially in critical decision areas related to loan approval or decisions on investments, is considered challenging (Brown et al., - 2020).

The financial services must also consider the growing challenges of how to align AI systems with regulatory compliance and ethical standards, since sometimes AI models evolve or adapt in ways that may not be predictable in all behaviors.

- Economic and Resource Constraints

Implementation of AI in banking is associated with a number of significant economic and resource-related challenges. High upfront costs for AI technologies, infrastructure upgrades, and maintenance may be beyond the capacity of small banks with modest budgets. In addition, a shortage of skilled personnel who can design, develop, and manage AI systems represents a notable barrier to the diffusion of AI (Thomas et al., - 2021)

It is a process that requires heavy investments in either training or acquiring specialized talents, which can be quite expensive for banks. The shortage of skilled professionals further affects the effectiveness in AI adoption since poor deployment or management results in poor system performance.

- Ethical and Legal Concerns

Ethical and legal issues are some of the major obstacles to the complete integration of AI into banking. Data privacy, especially regarding sensitive information about customers, raises very serious concerns as to how personal data is collected, stored, and used by the AI systems. This presents a challenge in the balancing act between offering personalized services and ensuring the privacy of users (Lee & Wong., - 2020).

Algorithmic bias can lead to cases where AI models favor one demographic over others, further increasing inequities in financial services, such as access to credit. Second, as AI become part of the equation, including job displacement and a concentration of power in large organizations with significant AI capabilities.

(Anon., n.d.), (Clifford G. Lau, 2018), (Broadbent, Meredith, 2021),

2.6. Critical Analysis on Empirical Gap

While the research on AI in banking is growing, there is still an empirical gap that warrants concern. Though the benefits, applications, and limitations of the technology have been considered, little has been done in terms of context-specific research to focus on how customers experience banking with AI-powered platforms, such as in Sri Lanka. This represents a huge knowledge gap in understanding how AI innovations play out in customer satisfaction and engagement, and how these vary across diverse cultural, economic, and technological landscapes.

- Lack of Context-Specific Research

Most of the available literature is focused on AI adoption in established markets like North America and Europe, with lesser emphasis on emerging markets like Sri Lanka. Various studies that take into consideration such unique characteristics as technological infrastructure, level of customer digital literacy, and relevant local regulatory frameworks are limited in number. Given that Araliya Bank operates in a context where digital adoption and AI technology are still evolving, the existing research does not fully capture the nuances of customer interaction with AI in such environments. (Brown et., - 2021)

- Limited Attention Paid to Customer-Centric Results

Whereas many studies exist in the literature with regard to technical applications, such as fraud detection or operational automation, very few studies purely investigate customer experience outcomes, and this is more so in Sri Lanka's perspective of banking. Empirical studies, while directly measuring the impact of AI tools on customer satisfaction, loyalty, and engagement, remain scant and largely unexplored in the correlating factors with tangible banking outcomes. (Lee and Wong., - 2020)

From a customer perspective, comprehensive research focused on how such AI tools in banking, ranging from personalized advice to chatbots or predictive analytics, would influence perceptions of trust and satisfaction remains necessary.

- Measuring AI's Effect on Customer Satisfaction Presents Challenges

Another gap exists in how the effectiveness of AI is measured in improving customer satisfaction. Most existing studies are based on either theoretical models or general consumer sentiment analysis without direct links to customer satisfaction related to measurable outcomes in the banking environment. The need for empirical research using direct data from customer-survey, interview, and transactional data-on the assessment of how AI systems in banks influence customers' decisions making processes, banking behaviors, and satisfaction in general. (Johson & Lee., – 2022).

- Empirical Studies on Prolonged Impacts

Another big gap is the long-term effects of AI on customer loyalty and retention within the banking framework. Much research has focused in short-term effects with regard to the customer reaction toward new platforms or products. How repeated AI interactions affect customers' trust, satisfaction, and loyalty overtime is something a bank should understand in order to establish relations with its customers for the long term. It would call for longitudinal research on customer behaviors in order to understand whether AI systems continue to meet evolving AI systems continue to meet evolving expectations or if there are diminishing returns over time (Green et al., 2021)

- Lack of Studies with Comparisons

There is a dearth of comparative analyses that review the application of artificial intelligence within different banking institutions or geographical regions. A study on the customer experience in using AI-driven platforms in Sri Lanka, in comparison with other South Asian or developing economies, would deliver valuable insights into how regional disparities affect the effectiveness of AI in the banking industry. Such comparative studies would help banks like Araliya Bank understand better the specific factors that driven or hinder customer satisfaction and assist in tailoring AI-powered platforms to meet local expectations more effectively. (Smith et al., - 2020)

3. CHAPTER THREE – METHODOLOGY

3.1. Introduction

This chapter presents the research methodology that has been used to study the practical applications, benefits, and limitations of Artificial Intelligence in enhancing customer experience at Araliya Bank's digital banking platform. The target population is described in this chapter, including the sampling techniques, data sources, study design, and instruments used to collect and analyze the data. The methodology is guided to the effect that the findings are valid, reliable, and agree with the objectives of the study. The chapter, therefore, lays the foundation for understanding the impact of AI-powered online transaction platforms on customer satisfaction and experience through a systematic approach.

3.2. Research Design

This research employed a mixed-method approach, combining both quantitative and qualitative methodologies to comprehensively understand the benefits and limitations of Araliya Bank's AI-powered online transaction platform in enhancing customer experience.

Quantitative Methods

Customer provided measurable data through the use of structured surveys with close-ended questions. The questionnaires targeted the degree of customer satisfaction, perception of AI technologies, and their usability in everyday banking interactions. Quantitative research has allowed the trends, patterns and statistical significance of insights to be spotted, providing a wide perspective on the effects that AI-driven digital banking features will have.

Qualitative Methods

Quantitative data was complemented by open-ended survey questions, which captured responses in detail and context from customers. These quantitative inputs provided deeper insights into customer expectations, preferences, and concerns that numerical data alone cannot capture. The approach gave deep insight into the customer perspective on the AI-powered platform.

Triangulation and Integration

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The mixed-method design has utilized triangulation in enhancing the validity and reliability of the findings through cross-validation of data from quantitative and qualitative sources. Thus, the combination of numerical data with narrative insights could therefore be made conduct a balanced analysis, posing both surface-level trends and deep causes of customer perceptions.

This thus helped the above-mentioned dual approach of the study to give a very sound framework in order to review the practical applicability, advantages, and limitations of AI in Araliya Bank's digital banking platform. It ensures that customers-centric insights are actionable and relevant to optimize digital banking services for overall customer satisfaction.

3.3. Sample

This is important for a representative and reliable sample. Thus, for the purpose of this research work, the calculated sample size shall consist of 20 customers in Araliya Bank, including heterogeneous classes on grounds of demography, geography, age and gender groups, and even economic class. Therefore, in order to catch a wide picture of experiences and perceptions related to customer artificial intelligence online-based transportation, it would be quite significant if different groups are captured. Respondent were then divide into two key groups,

- Heavy Users -: They constitute those customers who frequently use the digital banking platform for a wide range of transactions.
- Occasional Users -: Those who use the site infrequently or for specific purposes.

This segmentation allowed a comparative analysis of the experiences of users, highlighting variation in satisfaction, expectations, and challenges across user types. The sample size ensures statistical validity of the data without sacrificing the ability to pick up detailed insights about customer behavior and attitudes.

3.4. Sampling Procedure

The sampling procedure combined randomly and non-probability methods indrawing samples from a population of Araliya Bank customers who use the AI-powered online transaction

platform. In this way, the eventual sample was varied and representative, with participants being selected via a hybrid methodology that also did not violate ethics in research.

Random Sampling

A random sampling method was followed by in the selection of participants from the pool of existing customers actively using Araliya Bank's digital banking platform. This approach would reduce selection bias and increase the generalizability of the findings. The customers were randomly selected from a predefined list of users who had accessed AI-powered features where every eligible person had an equal chance of being chosen. This helped to reduce the danger of over-representing specific demographic ground and preserved the reliability of the results of this study.

Non-Probability Samplings

Non-Probability sampling was used to specifically access unique segments of the customer population, such as active users of the AI-powered platform, as it was impossible to get access to a large customer database. In the survey, customers were target via direct communication channels, including email invitations and WhatsApp messages, to request their participation. The participation was totally voluntary, and a link a Google Form was sent enable the customer to respond at their convenience.

This methodology supported adaptability and encouraged forthrightness by relieving the pressure on respondents. Even though non-probability sampling lacks proportional representation, it garnered critical information from regular users, who are also better acquainted with the inner workings of the application.

Combining both sampling method in this research created a holistic presentation of user experience and thus provided ample opportunity for exploring the pros and cons associated with Araliya Bank's AI-enabled digital banking application.

3.5. Data Sources

3.5.1. Primary Data

Data collection involves gathering information to find a solution or outcome to a specific question or problem. Research in academics might collect data to answer, for instance, the question of the prevalence of a certain disease in this or that particular region. A professional in international development, say, would otherwise collect data that show whether the program is truly effective or not. (Marta Costa , 2022)'

Customer Feedback

A structured Google Form survey will be the primary data collection source for this research, designed to collect insight from Araliya Bank customers. The survey aims at covering experiences of customers in using the AI-powered online transactions platform and their level of satisfaction. Key aspects covered in the survey include,

- Ease of Use -: How easy it is to use the platforms who have different levels of technological proficiency
- Efficiency -: The speed and how well the transaction goes through, including problems that were encountered in the process of transacting.
- Reliability -: Feedback on the consistency of the performance of the platform in terms of uptime and responsiveness.
- Personalization -: How well the platform personalizes its services to meet the needs of each customer, such as customized recommendations or notifications.
- Overall Satisfaction -: The general feeling of the platform, whether customers would recommend the service to others.

The survey will target only the active Araliya Bank customers who have used the digital platform in order to ensure that data collected is relevant and reliable. Quantitative analysis will be done using rating and scale-based questions, while the qualitative part relates to open-ended responses for determining the strengths and weaknesses of the AI-powered platform.

3.5.2. Secondary Data

Secondary data is defined as information gathered by anyone other than the primary data. Typical sources of secondary data in the social science include sinuses, data collected by government agencies, organizational records, and data originally collected for the other research purposes. On the other hand, primary data is directly collected by the researcher who is carrying out the study. (Contributors, 2019)

The research on Araliya Bank’s AI-powered online transaction platform will be supported by the following secondary data sources,

Araliya Bank 2021 Stakeholder Report

The technical innovations of Araliya Bank, such as their AI-powered services, are discussed in this paper along with how they relate to customer happiness. (PLC, 2021)

Araliya Bank 2023 Annual Report

An updated summary of the bank’s digital projects, performance, and customer experience strategies particularly with regard to artificial intelligence is given in this paper. (Anon., n.d.)

Connecting with Our Stakeholders 2024

Araliya Bank’s stakeholder engagement initiatives are included in this publication, along with comments on customer satisfaction and digital banking services. (Anon., n.d.)

CSE Report on Araliya Bank

Araliya Bank’s financial and strategic analysis is included in this report, along with information on their use of AI and digital banking systems. (Anon., n.d.)

Araliya Bank Wikipedia

Araliya Bank, details related to history, growth, technology, and some other details taken from this website. (Contributors, 2019)

3.6. Data Collection Instrument / Survey Instrument

- **Survey Tools**

Platforms like Google Forms offer user-friendly interfaces for creating, distributing, and analyzing survey responses, allowing Araliya Bank to efficiently gather information from a large audience.

- **Data Recording Tools**

Tools for recording observations or interactions, such as voice recorders or note-taking apps, aid in capturing details during on-site visits or direct observations.

- **Text Analysis Tools**

Platforms like NVivo, Prevalis Research, or Leximancer assist in qualitative analysis, helping Araliya Bank extract meaningful insights from textual data obtained from open-ended survey responses or interviews.

- **Data Visualization Tools**

Visualization tools like Tableau, Power BI, or Google Data Studio create visual representations of analysed data, making it easier for Araliya Bank to present complex information in an understandable format.

06/01/2025, 01:15

Araliya Bank Sri Lanka

Araliya Bank Sri Lanka

We'd love to hear your feedback on how we can make your experience even better! 🌟

* Indicates required question

Araliya Bank Customer Feedback Form

Dear Participants,

We invite you to participate in a survey aimed at understanding the benefits and limitations of using Araliya Bank's AI-powered online transaction platform to enhance customer experience and satisfaction. Your insights are invaluable in identifying areas where our digital banking services can better serve your needs, improve convenience, and provide greater satisfaction.

This survey focuses on your usage and knowledge of Araliya Bank's digital banking platform, your perceptions of its strengths and challenges, and your feedback on how we can enhance its features to deliver a superior customer experience.

Your participation is voluntary, and your responses will remain anonymous. Your valuable input will contribute to our ongoing efforts to innovate and improve our digital services, ensuring we remain a leader in customer-focused banking solutions.

Thank you for your time and support in helping us serve you better.

Sincerely,

Project Manager,

Araliya Bank(Sri Lanka)

1. **01. Name ***

2. **02. User Details ***

Mark only one oval.

- ☐ Technical
- ☐ Non-technical

3. **03. What is your age group? ***

Mark only one oval.

- ☐ Under 18
- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45+

https://docs.google.com/forms/d/1f-ALs23dIM2QEC23h1278cJ_QXegg1dphKrHcSoBnB-g/edit

1/8

Figure 8 - Feedback Form

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4. 04. What is your gender? *

Mark only one oval.

☐ Male

☐ Female

5. 05. How often do you use Araliya Bank's digital banking platform? *

Mark only one oval.

☐ Daily

☐ Weekly

☐ Monthly

☐ Rarely

Untitled section

Choose your Feedback

6. 06. How satisfied are you with the overall platform? *

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

7. 07. How secure do you feel while conducting transactions on the platform? *

Mark only one oval.

☐ Very secure

☐ Somewhat secure

☐ Neutral

☐ Somewhat insecure

☐ Very insecure

Figure 9 - Feedback Form

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8. **08. Which feature of the platform do you use the most? ***

Mark only one oval.

☐ Fund transfers

☐ Bill payments

☐ Account balance inquiry

☐ AI chatbot support

9. **09. How would you rate the speed of the platform? ***

Mark only one oval.

1 2 3 4 5

Poo ☐ ☐ ☐ ☐ ☐ Excellent

10. **10. How reliable is the platform during peak hours? ***

Mark only one oval.

☐ Always reliable

☐ Mostly reliable

☐ Sometimes reliable

☐ Rarely reliable

11. **11. How intuitive is the platform's user interface? ***

Mark only one oval.

☐ Very intuitive

☐ Moderately intuitive

☐ Neutral

☐ Not intuitive

12. **12. Do you find the AI chatbot helpful in resolving your queries? ***

Mark only one oval.

☐ Yes

☐ No

☐ Maybe

https://docs.google.com/forms/d/1f-ALs23dM2QEC23h1278cJ_QXegg1dphKrHcSoBnB-g/edit 3/8

Figure 10 - Feedback Form

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13. **13. How often do you face technical issues with the platform? ***

Mark only one oval.

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

14. **14. Would you recommend the platform to others? ***

Mark only one oval.

☐ Yes

☐ No

☐ Maybe

15. **15. How confident are you in the bank's efforts to improve the platform? ***

Mark only one oval.

☐ Very confident

☐ Somewhat confident

☐ Neutral

☐ Not confident

16. **16. Does the platform offer enough features for your banking needs? ***

Mark only one oval.

☐ Yes, it offers all necessary features

☐ Yes, but some features could be added

☐ No, it lacks some important features

☐ No, it's inadequate

Figure 11 - Feedback Form

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17. 17. How personalized do you feel your banking experience is on the platform? *

Mark only one oval.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ Highly personalized

18. 18. How does the platform compare to other digital banking services you've used? *

Mark only one oval.

☐ Much better

☐ Slightly better

☐ About the same

☐ Worse

19. 19. Do you believe the platform saves you time compared to traditional banking? *

Mark only one oval.

☐ Always

☐ Mostly

☐ Sometimes

☐ Rarely

Untitled section

Give your Feedback

20. 20.What specific features do you find most beneficial on the platform? *

<https://docs.google.com/forms/d/1fALs23dM2QEC23h1278eLQYee1dobKkleSoRnBn/edit> 5/8

Figure 12 - Feedback Form

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06/01/2025, 01:15 Araliya Bank Sri Lanka

21. 21. Can you describe any challenges or issues you've faced while using the platform? *

22. 22. What additional features would you like to see added to the platform? *

23. 23. How has the platform impacted your overall banking experience? *

24. 24. What do you consider the platform's biggest limitation? *

25. 25. What suggestions do you have for improving the AI-powered features of the platform? *

https://docs.google.com/forms/d/1f-AI_s23dM2OEC23h1278cJ_QXegp1dphKrHcSoRnB-a/edit 6/8

Figure 13 – Feedback Form

06/01/2025, 01:15 Araliya Bank Sri Lanka

26. 26. What motivates you to continue using Araliya Bank's digital platform? *

This content is neither created nor endorsed by Google.

Google Forms

Figure 14 - Feedback Form

3.7. Development of the Questionnaire

The development of a survey instrument to measure customers' experiences with Araliya Bank's AI-driven online transaction platform requires careful planning of the survey to gather both demographic data and data based on objective criteria. The questionnaire design has to be tailored to the specific research objectives of explaining the impact of AI on customer experience and satisfaction, including the benefits and limitations of the platform.

The framework and justification for creating the questionnaire are provided below,

3.7.1. Demographic Section

Customer segmentation (age, gender, frequency of usage, etc.) and the ways in which various groups see and utilize the AI-powered online transaction platform will be revealed by demographic data.

Important questions in the demographic section,

- **Age** -: To understand how users of all ages interact with the platform
- **Gender** -: To identify whether user experience varies according to gender
- **Frequency of Use** -: To determine how frequently users engage with the platform, since this may influence how satisfied and functional they find it.

3.7.2. Questions to Access Research Objectives

The main purpose of this research is to identify the benefits and limitations involved in Araliya Bank's AI-driven online transaction platform for improving customer experience. The survey instrument is designed to gather information on how the system fares in terms of usability, security, personalization, customer support, and overall satisfaction; it should also look into possible limitations, including technical difficulties and issues related to data privacy.

Objective 01 -: User Satisfaction and Experience

Questions,

- “How satisfied are you with the overall platform?”
- “How intuitive is the platform’s user interface?”
- “How secure do you feel while conducting transactions on the platform?”

This objective is gauge how satisfied user are with the platform’s security, dependability, speed, and general features utilization. It aids in determining how pleased customers are with Araliya Bank’s digital platform’s usability and functionality.

Objective 02 -: Platform Performance and Trustworthiness

Questions,

- “Which feature of the platform do you use the most?”
- “How reliable is the platform during peak hours?”
- “How confident are you in the bank’s efforts to improve the platform?”

This objective is to compare the platform to other online banking services, assess how well it performs during peak hours, and gauge customer trust. It aids in assessing the platform’s dependability, especially during periods of high demand, and if clients have faith in the bank’s attempts to enhance the service.

Objective 03 -: Feature Requests and Improvement Suggestions

Questions,

- “What specific features do you find most beneficial on the platform?”
- “Can you describe any challenges or issues you’ve faced while using the platform?”
- “What additional features would you like to see added to the platform?”

This objective is to gather opinions on the platform’s features, user difficulties, and ideas for future features. It aims to learn what users would like to have added to or improved upon, as well as how the platform affected their banking practices.

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Structuring the Questionnaire

- Question Types (Likert Scale 1-5, Multiple Choice, open-ended)
- Flow and Logic

3.8. Reliability and Validity of Questions and Responses

3.8.1. Reliability

Reliability refers to the consistency or stability of the responses to the questions across different instances and groups. In ensuring reliability within your research on Araliya Bank's AI-powered online transaction platform, the questions should be drafted in such a way that if the same conditions are maintained at different times, the results must be consistent. In ensuring reliability, several steps are necessary,

- **Clear and precise Wording** -: To prevent misunderstanding and guarantee that all responses understand the questions, they must be precisely.
- **Pilot Testing** -: Finding any problems with questions clarity or answer consistency can be facilitated by doing a pre-test or pilot survey with a small group.
- **Standardized Response Options** -: Consistent replies from a range of client can be obtained by using multiple choice questions or a Likert scale.

Validity

Validity refers to the precision with which the questions measure what is to be measured. In your research context, that would be questions that accurately capture the experience and satisfaction of customers on Araliya Bank's AI-driven platform. Types of validity include,

- **Content Validity** -: Making sure the enquiries thoroughly address every relevant facet of the client experience, including usability, security, dependability, and satisfaction.

e.g., Enquiries concerning user interface, security, and transaction speed should cover every important aspect of the client's engagement with the platform.

- **Construct Validity** -: Confirming that the questions accurately assess the fundamental concepts of customer experience and satisfaction. This may be

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accomplished by matching the queries to accepted frameworks and theories of digital banking customer experience.

e.g., The motion of satisfaction may be measured with the inclusion of questions about perceived convenience, trust, and personalization.

- **Criterion Validity** -: To determine if the survey results are consistent with actual behaviors and experiences, this may be evaluated by contrasting the replies to predetermined benchmarks or objectives.

e.g., Verifying the legitimacy of survey relies on can aided by comparing satisfaction ratings with platform usage statistics.

4. CHAPTER FOUR – DATA ANALYSIS AND RESULTS

4.1. Introduction

This chapter present an analysis of the data collected on evaluating the benefits and limitations of Araliya Bank’s AI-powered online transaction platform in improving customer experience and satisfaction. The chapter first gives an overview of the data collection and tools used for the analysis, followed by the demographic profile of the respondents, and then a critical evaluation of the survey results. Key findings have been clarified under benefits and limitations supported by univariate and inferential satisfaction analysis. The chapter, therefore, concludes with the trends and patterns governing the research objectives to form a basis for discussion and recommendations in the next chapter.

4.2. Descriptive Data Analysis on Demographic Information

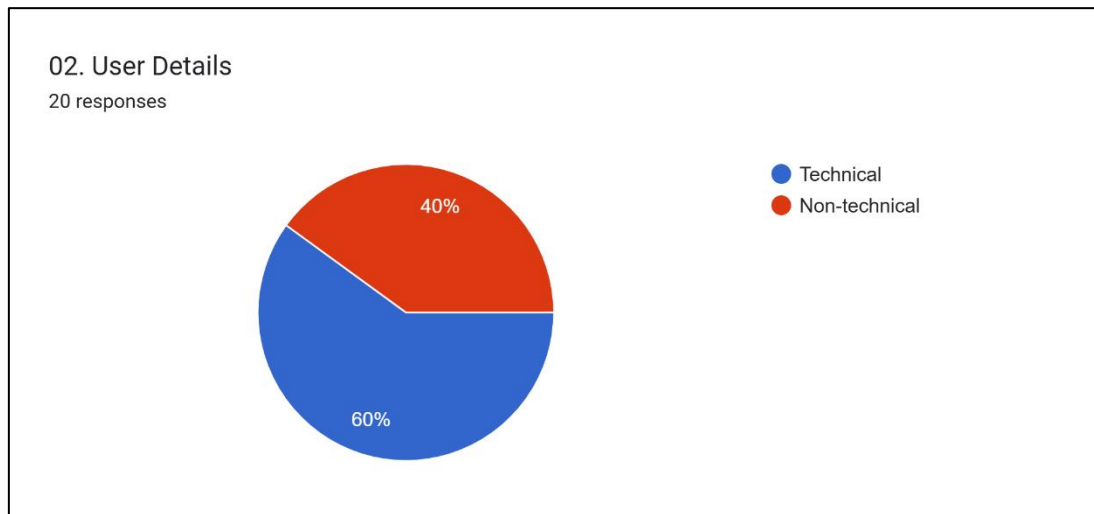


Figure 15– Demographic Information

The User Details distribution indicate a diverse sample of, with the majority of respondents are technical users.

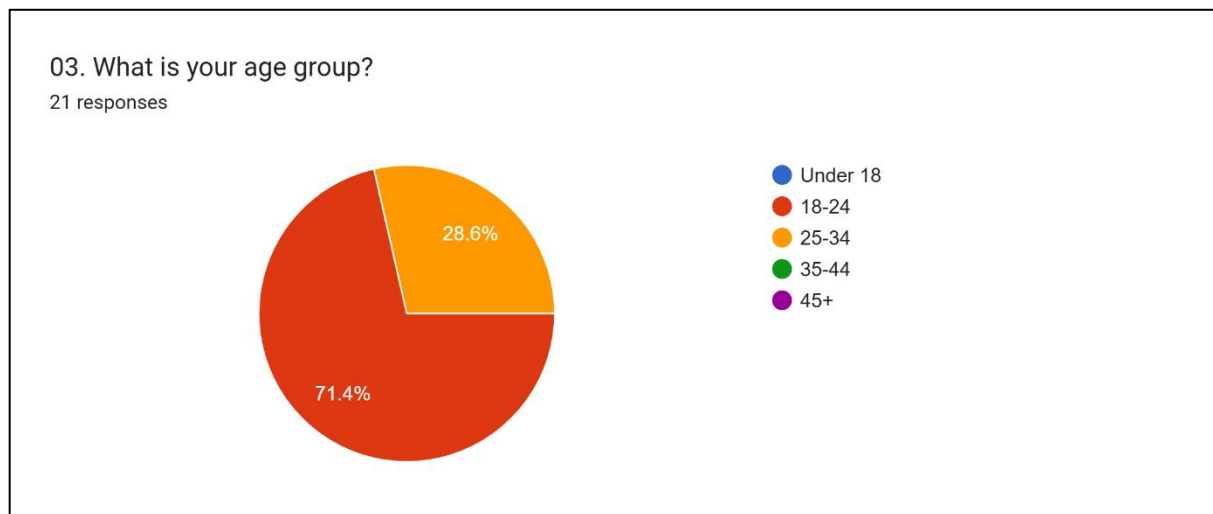


Figure 16 – Demographic Information

The age distribution indicates a diverse sample, with the majority of respondents in the 18-25 age range, reflecting a workforce that is relatively young and likely more adaptable to new technologies.

04. What is your gender?

21 responses

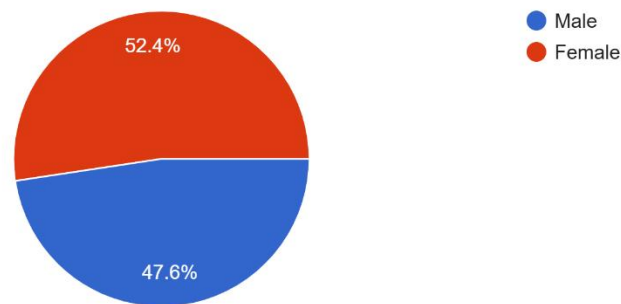


Figure 17 – Demographic Information

The gender distribution shows a balanced representation, with a slight majority of female respondents. This balance is essential for understanding diverse perspectives on AI implementation.

05. How often do you use Araliya Bank's digital banking platform?

21 responses

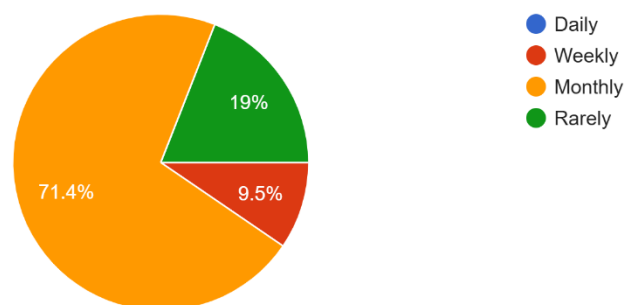


Figure 18 – Demographic Information

The pie chart shows majority of users access the Araliya Bank Digital Banking platform monthly, and least percentage of users access daily.

4.3. Descriptive Data Analysis on other variables to access objectives

Objective 01 -: User Satisfaction and Experience

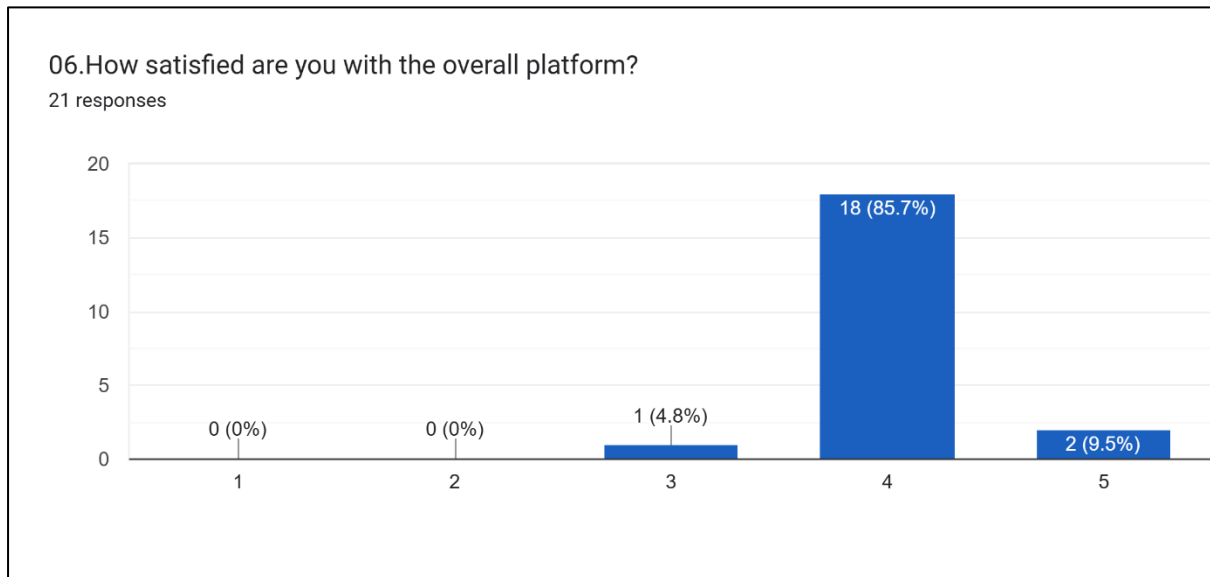


Figure 19 - User Satisfaction and Experience

The bar graph shows the satisfaction levels with Araliya Bank's digital banking platform, majority rated satisfaction as 4 stars, indicating overall positive feedback.

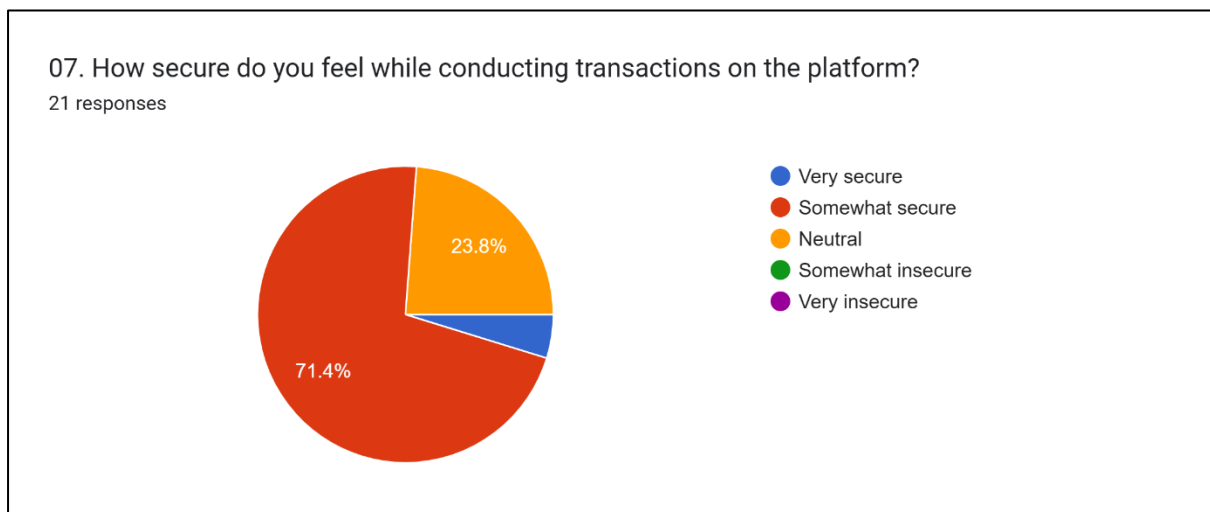


Figure 20 – User Satisfaction and Experience

This pie chart shows the majority feel somewhat secure about the security of the platform and least percentage about respondents feel very secure about Araliya Bank digital banking platform.

Objective 02 -: Platform Performance and Trustworthiness

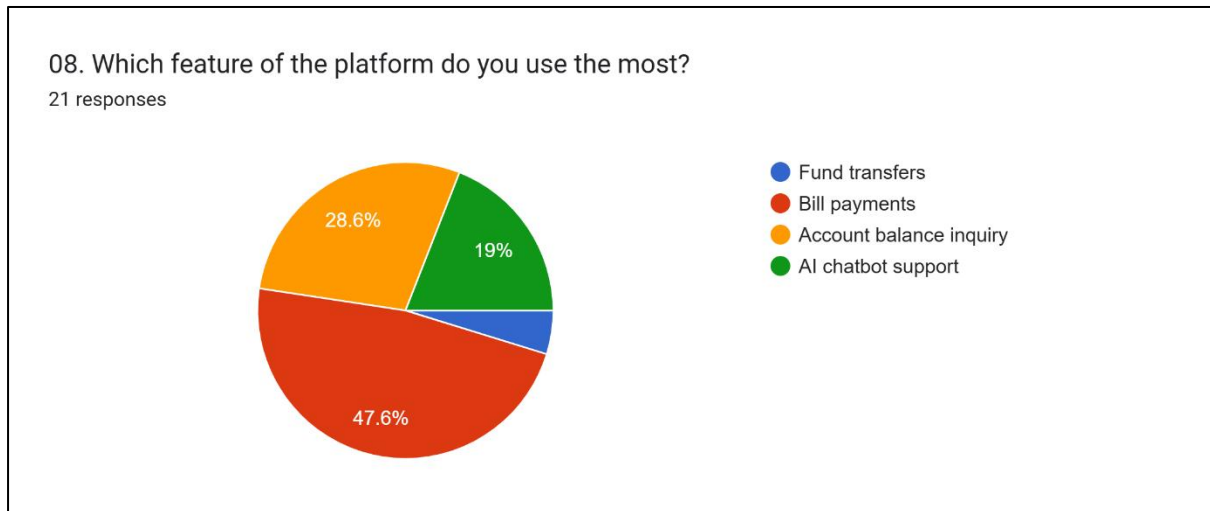


Figure 21 - Platform Performance and Trustworthiness

The chart highlights the majority percentage of respondents use the platform for bill payment, followed by account balances inquiries.

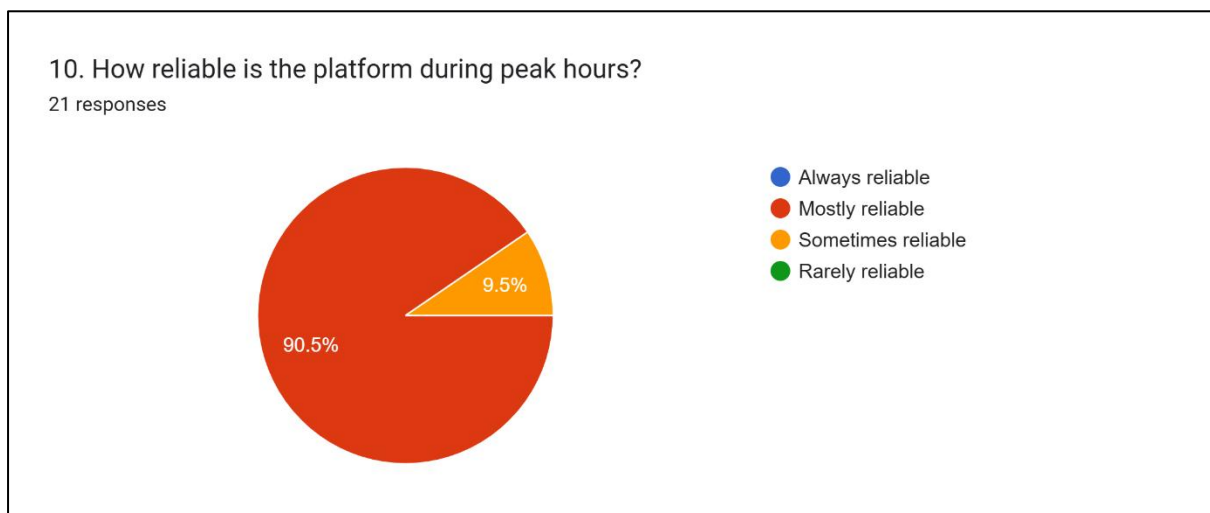


Figure 22– Platform Performance and Trustworthiness

This chart shows the platform's reliability during peak hours, the majority percentages of respondents mostly reliable during peak hours, with a small portion rating it as always reliable.

Objective 03 -: Feature Requests and Improvement Suggestions

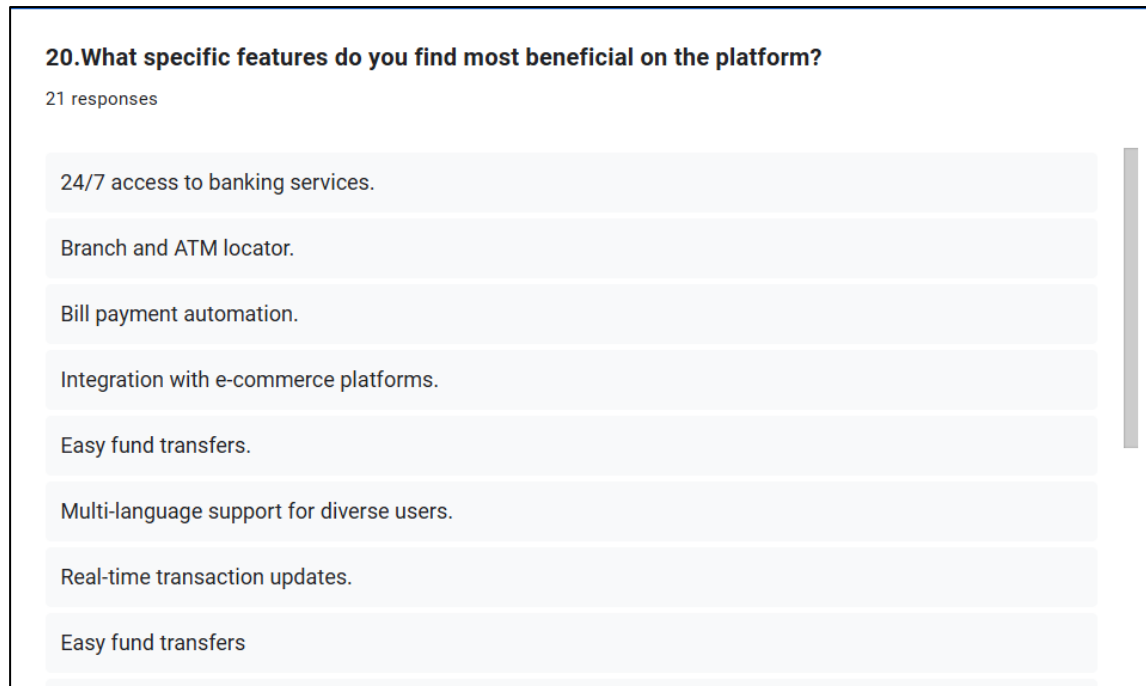


Figure 23- Feature Requests and Improvement Suggestions

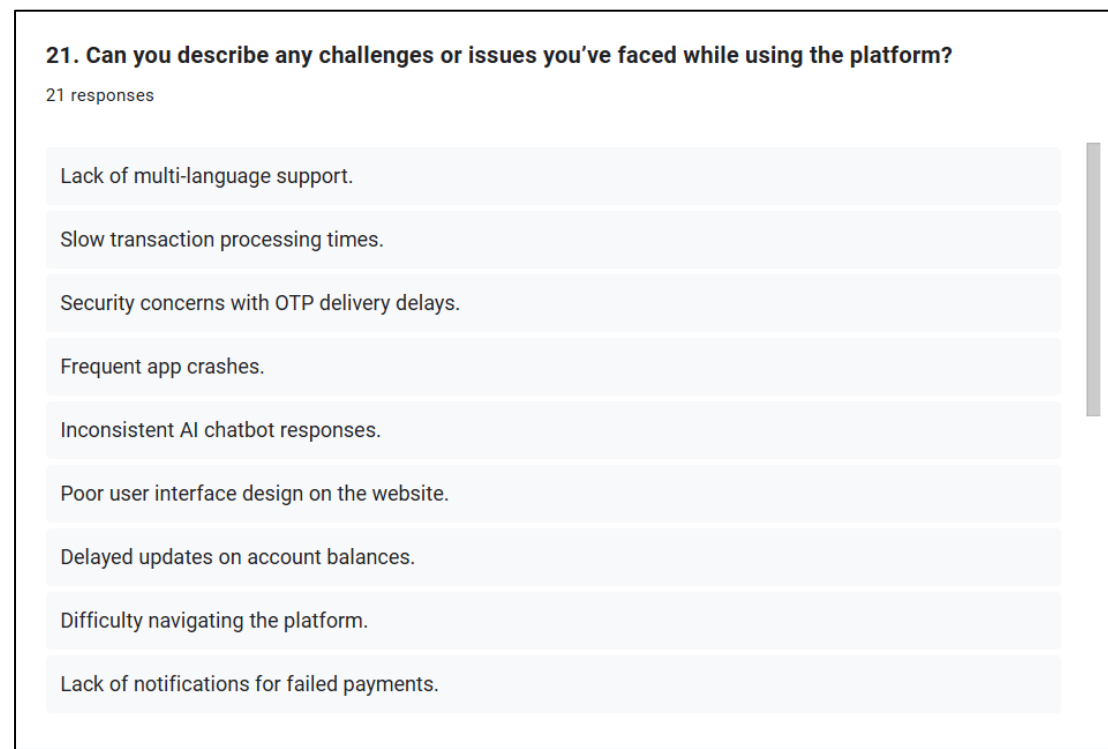


Figure 24- Feature Requests and Improvement Suggestions

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5. CHAPTER FIVE -CONCLUSION AND RECOMMENDATIONS

5.1. Introduction

This chapter summarizes an overview of the findings from the research on Araliya Bank's AI-powered online transaction platform and its impact on customer experience and satisfaction. The benefits and limitations that the platform carries with itself to improve the overall customer experience were the main aspects this study aimed to evaluate. This is a summary chapter where the conclusions developed from the data collected through the survey among customers and secondary research are summarized. It also provides recommendations on how best to enhance the functionality of the platform, resolve challenges, and optimize customer experience. It is expected to provide useful input to optimize customer experience. It is expected to provide useful input to Araliya Bank in its quest for continuous evaluation and survival in the highly competitive, dynamically challenging digital banking environment.

5.2. Summary of Findings

The main points of this research work were the evaluation of the impact of Araliya Bank's AI-powered online transaction platform on customers' experiences and satisfaction, benefits, and limitations. The research tried to explore the contribution of the platform to enhance customer experience, operational efficiency, security, and access to services and also identify those areas where the platform needs further improvement.

Brief Finding

- **Improved Customer Experience** -: The AI-powered enhances customer experience through personalization of services and timely support. Indeed, several customers appreciated the fact that recommendations were more personalized, with response time faster, hence increasing customer satisfaction levels.
- **Operational Efficiency and Convenience** -: The efficiency and speed at which the platform execute transactions, added to the fact that it is accessible to 24/7, increase convenience for customers. Several users noted the ease of conducting transactions from anywhere at any time.

- **Security and Fraud Prevention** -: AI-powered security features, such as fraud detection and risk analysis, were most valued. It gave customers more confidence in the security of their transactions and thus had a great impact on customers' trust in the platform.
- **Limitations and Challenges** -: Although the above-mentioned list of benefits, a number of limitations were apparent. Technical glitches or slowness with clients occurred at times and, at times, delayed the transaction process itself. There were some apprehensions linked to data privacy and no interaction with human personnel in areas such as customer service.
- **Customer Trust and Adoption** -: Even as the attained remarkable reach with high adoption, it still had to work on the understanding or acceptance of the AI technology because some users found it quite impersonal, others somewhat scared. It important that education be provided about the capability and benefits of the platform.

The results indicate that although Araliya Bank's AI powered platform has many advantages, resolving the noted drawbacks might improve client satisfaction even further and help the company keep a competitive edge in the online banking market.

5.3. Recommendations

The research's conclusions lead to the following suggestions for enhancing Araliya Bank's AI-powered online transaction platform, enhancing customer satisfaction, and resolving the constraints found,

- **Improve AI Awareness and Education** -: Araliya Bank should therefore finance educational campaigns to reduce customer uncertainty about AI technology. These can include tutorials, FAQs, and even in app guidance about the benefits, security measures, and ease of use for the AI-enabled platform. Cleary communicating how AI work and where its advantages lie would results in increasing confidence and thus overall adoption by customers.

- **Enhancing Technical Support and Reliability** -: Generally, though the platform is efficient, there have been occasional technical glitches. Hence, Araliya Bank is supposed to enhance the backend infrastructure to ensure smooth transaction processing. Besides this, real-time support through AI chatbots or a dedicated helpline will provide customers with on the spot assistance in case any technical problems.
- **Improve Data Privacy and Security Arrangements** -: As much as customers like the security features, they express their concern on data privacy. Therefore, Araliya Bank needs to keep on investing in cutting edge security technologies like encryption and multi-factor authentication. Also, ensure alignment with international standards concerning protection of data. Clear transparency should be ensured as to how a customer's data is dealt with and kept.
- **Integrate Hybrid Customer Support** -: While AI-driven customer service offers speed and convenience, it may lack the personal touch needed in certain scenarios, for better customers satisfaction, Araliya Bank needs to integrate hybrid support system that involve a combination of AI with human customer service agents in complex inquiries or cases where emotional intelligence is required. This will help in providing full-circle customer support.
- **Continuous Improvement of the Platform, Inclusion of Feedback** -: Customer feedback is always necessary for continuous improvement. Araliya Bank should have a mechanism for continuously collecting user feedback regarding the features and performance of the platform. Based on the feedback, periodic updates should be released to fine-tune the functionality of the AI, user interface, and security features of the platform to maintain alignment with customer needs and expectations.

5.4. Limitations

To put the results and suggestions in perspective, it is important to recognize the following limits and restriction that this study on Araliya Bank's AI -powered online transaction platform faced,

- **Limited Scope of primary Data** -: Primary data for study came from customer surveys. The sample size and demographics diversity of respondents may not be fully representative of the broader customer base of Araliya Bank, which limits the generalizability of findings.
- **Dependence on secondary data** -: A large part of the research relied on secondary data sourced from the annual reports and other official documents of Araliya Bank. These sources, though reliable, may not possess the fineness of details to delve into certain specific in studying customer experience and satisfaction.
- **Technological Complexity** -: While trying to understand the technical functionalities of the AI-powered platform, there has been extensive secondary research that might not capture the complete depth of operational mechanisms. This could affect certain technical insights for accuracy.
- **Dynamic Nature of Digital Banking** -: The digital banking landscape continuously changes with continuous development in AI and changing customer expectations. The results of this study may be valid for a short period as new features and technologies are introduced
- **Customer bias in feedback** -: In may cases, customers' feedback from the survey individual biases and recent experience from the platform; positive and negative feedback does not constitute the overall general performance of a long team operating AI-powered system

5.5. Implications for Future Research

This study into the AI-driven online transaction platform of Araliya Bank sheds light on customer experience and satisfaction. On the other hand, it also shoes some areas that, if researched, could give further depth and bridge in the present study. The key implications for future research are listed below,

- **Comparative studies across bank** -: More studies can be made in the future to compare Araliya Bank 's AI powered platform with that of other banks, both locally and internationally, for industry best practices, performance benchmarking, and unique differentiator.
- **Longitudinal Studies** -: It would be more useful to conduct longitudinal study wherein customer satisfaction and the performance of the platform could be tracked over time to indicate how well it has adapted in the long term to evolving customer needs and technological change.
- **Focus on customer segments** -: Further research in this regard could be done on the differential impacts that the AI-powered platform has created across segments, such as SMEs, retail customers, and high-net-worth individuals, to understand how diverse customers needs are being catered to and further customize offerings.
- **Exploration of Advanced AI capabilities** -: In future studies, with the constant development in field of AI, the contribution of advanced AI features, such as predictive analytics, conversational AI, and machine learning, may be pursued in enhancing customer engagement and satisfaction in digital banking.
- **Behavioral Analysis of Adoption** -: Such could be deepened in analyzing customers' behavior, focusing on drivers and barriers for the adoption or refusal of an AI - powered platform, meaning perceptions , preferences, and concerns regarding AI in banking.

Activity 3 – Action Plan

1. Introduction

In this regard, an effective action plan is important for the AI-powered online transaction platform of Araliya Bank in order to maximize customer satisfaction and deliver a great user experience. This will ensure that the benefits and limitations identified are addressed, and innovation is fostered in line with customer expectations. The key objective is to make the best use of AI technology to enhance efficiency, security, and personalization while minimizing challenges in terms of data privacy and other accessibility barriers.

The action plan will follow a phased approach, starting with an overall assessment of customer feedback obtained through direct surveys and secondary data analysis. It will also focus on enhancement of the features of the platform, filling the gaps that have been identified, and introducing targeted improvements. Combining advanced AI capabilities with customer-centric strategies, the plan aims at strengthening the digital banking experience and building trust ensuring the platform remains adaptable to future technological advancements and market dynamics.

2. Plan to communicate project recommendations

To make sure that all stakeholders are aware of, involved with, and supportive of the suggested changes, it is imperative that project recommendations are communicated. The delivery methods, target audience, facilitators, and timeframes for efficiently distributing the recommendations are described in the plan that follows.

Action No	Recommendation	Delivery Mode	Target Audience	Facilitator/s	Dates	Time Period
1	Improve customer awareness regarding AI-powered platform features.	Physical Training	Customer Relationship Officers (CRO)	Internal Training Team	2024/12/19 – 2024/12/21	3 Days
2	Organize workshop on how to improve the user experience on the platform	Workshop	Digital Banking Team and IT Employees	External Consultant	2024/12/22/ - 2024/12/23	2 Days
3	Use AI technologies to train employees on security and fraud detection.	Online Seminar	All Employees	Internal IT Team	2024/12/24 – 2024/12/26	3 Days
4	Teach staff how to handle user input so that the platform may be Improved.	Physical Workshop	Customer Support Team	Internal HR Team	2024/12/27 – 2024/12/29	3 Days
5	Expand the platform's AI powered personalisation capabilities.	Online Workshop	Market and Product Development Team	Foreign Consultant	2024/12/30– 2025/01/01	3 Days

Table 6 – Planning to communicate project recommendations

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2.1. Materials to Communicate Research Project Outcomes

A variety of materials will be created and distributed in order to successfully convey the research project's findings to audiences that are both technical and non-technical. All stakeholders will be able to access, comprehend and implement the suggestions thanks to these resources.

- **Presentation** – Detailed PowerPoint presentations to showcase the conclusions, recommendations, and future steps of the research. Training classes, workshops, seminars, and conference will be facilitated by these presentations. They will ease the understanding of the stakeholders by offering a structured and visually attractive overview of the major conclusions and suggestions.
- **Policy Documents** – It will also comprise policy documents on best practice and ethical standards for the use of AI. These materials will be provided to all staff members and will also be available through the business internet. They shall act as resources to ensure that AI is used ethically and in an open manner.
- **Wallpaper** – Informative infographic and wallpaper that outline major study conclusions and recommendations. To create greater awareness and to educate customers of the bank, regarding the AI work and its benefits, these graphics will be placed in common spaces of the firm.
- **Posters** – Colorful posters announcing the upcoming training sessions and events while emphasizing the benefits of AI technologies. Placing posters carefully in high traffic area within the bank, such as the hallways and break rooms, is going to help capture the attention of the employees and foster participation.
- **User Manuals** – Detailed user manuals and instructions for the specific AI systems that Araliya Bank customers. These will give complete directions on how to effectively use AI technologies and apps, according to industry best practices. When using AI technology, these will prove to be valuable resources for employees.

3. Present arguments for the planning decisions made when developing the project plans

To guarantee the project's success, a number of crucial planning choices were made when creating the project plans for Araliya Bank's AI implementation. The following are the justifications for every planning choice.

	Planning Decision Made	Present Argument
1	Define the Objectives and Scope Definition to guide the project to meet the expectations of the organization.	Research objectives were set to align with the Araliya Bank objective of using AI-driven digital banking services to enhance customer satisfaction. The scope was clearly defined to focus on the analysis of the benefits and limitations without deviation from the core topic.
2	Undertake a serious risk assessment to single out potential challenges and uncertainties that might jeopardize the success of the research project.	Among others, risks relating to data availability, stakeholder participation, and technology barriers were identified. Then mitigation measures for those risks were charted through a detailed risk management plan tailor-made for the banking industry.
3	Proper resource allocation and budgeting to ensure that the project has the necessary resources to achieve its objectives within the stipulated time frame and budget.	Resources were optimized, especially in term of customer survey instruments, secondary data sources, and expert consultations. A tight control over the budget ensured that no financial overruns occurred while meeting the project goals effectively.
4	Develop a realistic project timeline with clear milestones that will help track progress, identify bottlenecks,	Some of the activities that were involved in the timeline of the project included survey distribution, data collection, analysis, and reporting. This ensured that research was

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	and ensure the project stays on schedule.	complete within the stipulated time while making adjustments where needed.
5	Throughout the project, involve stakeholders and maintain efficient communication to guarantee that the requirements and expectations of all parties are met.	The insights were most valuable, coming from regular communication with the stakeholders of Arliya Bank, including management, IT teams, and customers. Relevant and actionable insights for improving customer experience were delivered through workshops and presentations.
6	Apply appropriate quality standards to ensure that the project will meet high-quality requirements.	Industry standards ensured that improved to the chatbots conformed to high-quality benchmark standards.

Table 7 – Present arguments for the planning decisions made when developing the project plans

3.1. Changes of the Initial Project Management Plan

3.1.1. Revised Budget

Here is an updated budget that reflects possible modifications as a result of the project management plan revisions to convey the suggestions, as shown by the cost calculation above,

- **Training and Development**

It is increased by LKR 5000 to cover the added sessions of new methods and tools of communication.

- **Operation Costs (Supplied)**

LKR 3000 increased for the extended use of office supplies by the project in question.

- **Reporting and Documentation (Printing)**

Increased by LKR 2000 to cater to more extensive reporting and printing requirements.

- **Emergency Fund**

Changed to reflect 10% of the updates total budget.

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Category	Cost Items	Description	Revised Estimated Costs (LKR)
Personnel	Project Manger	Pay for the duration of the project	80, 000
	Technical Team	Team member salaries (AI, ML, DA, etc.)	87, 000
	Research Team	Research and data collector salaries	75, 000
	Administrative Team	Administrative assistance salary	66, 000
Software and Equipment	Software Licenses	AI/ML software, AI tools	60, 000
	Hardware	Computers, Servers, and Storage devices	62, 000
	Cybersecurity Tools	Software for data protection	57, 000
Survey Tools	Online survey Platforms	Survey tools subscription fees	12, 000
Data Collection	Internet	Secondary data purchase	20, 000
	Travelling	Stakeholder meetings and data collection travel costs	30, 000
Training and Development	Seminars and workshops	Team training on new technology	25, 000 (Increased)

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Operation Costs	Supplies for the office	Standard office supplies	15, 000 (Increased)
	Communication	Phones, Internet, and other communication tools	25, 000
Reporting and Documentation	Production of report	Report printing and binding	12, 000 (Increased)
	Presentation resources	Making resources for presentations	18, 000 (Increased)
Emergency	Emergency fun	10% of the overall budget is set aside for extra expenses	70, 000 (Changed to reflect 10% updated total budget)
Total Revised Budget			710, 000

Table 8 – Revised Budget

3.1.2. Revised Milestone Chart/ Gannt Chart

Milestone	Initial Timeline	Revised Timeline	Change Reason
Project Charter Development	2024/11/01 – 2024/11/03	-	No change
Feasibility Study	2024/11/04 – 2024/11/07	-	No Change
Initial Risk Assessment	2024/11/08 – 2024/11/10	-	No Change
Project Plan	2024/11/11 – 2024/11/17	-	No Change
Research Work	2024/11/18 – 2024/11/23	-	No Change
Process Completion	2024/11/23 – 2024/11/25	-	No Change
Data Collection	2024/11/26 – 2024/12/05	-	No Change
Data Analysis	2024/12/06 – 2024/12/18	-	No Change
Improve customer awareness regarding AI-powered platform features.	-	2024/12/19 – 2024/12/21	Added to raise consumer awareness of platform benefits driven by AI
Organize workshop on how to improve the user experience on the platform	-	2024/12/22 – 2024/12/23	Added to provide user experience design training to the IT staff and the digital banking team
Use AI technologies to train employees on security and fraud detection.	-	2024/12/24 – 2024/12/26	Added to teach staff members how to use AI technology for security and fraud detection

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Teach staff how to handle user input so that the platform may be Improved.	-	2024/12/27 – 2024/12/29	Included to instruct staff how to manage user feedback for platform enhancement
Expand the platform's AI powered personalisation capabilities.	-	2024/12/30 – 2025/01/01	Added to improve external teams' personalisation capabilities driven by AI
Final Review and Evaluation	2024/12/19 – 2024/12/22	2025/01/02 – 2025/01/05	Postponed allowing for further communication actions
Documentation and Knowledge Transfer	2024/12/23 – 2024/12/25	2025/01/06 – 2025/01/08	Postponed allowing for further communication actions
Project Handover	2024/12/26 – 2024/12/28	2025/01/09 – 2025/01/11	Posted allowing for further communication actions
Project Closure Report	2024/12/29 – 2024/12/31	2025/01/12 – 2025/01/14	Posted allowing for further communication actions

Table 9 – Revised Milestone Chart

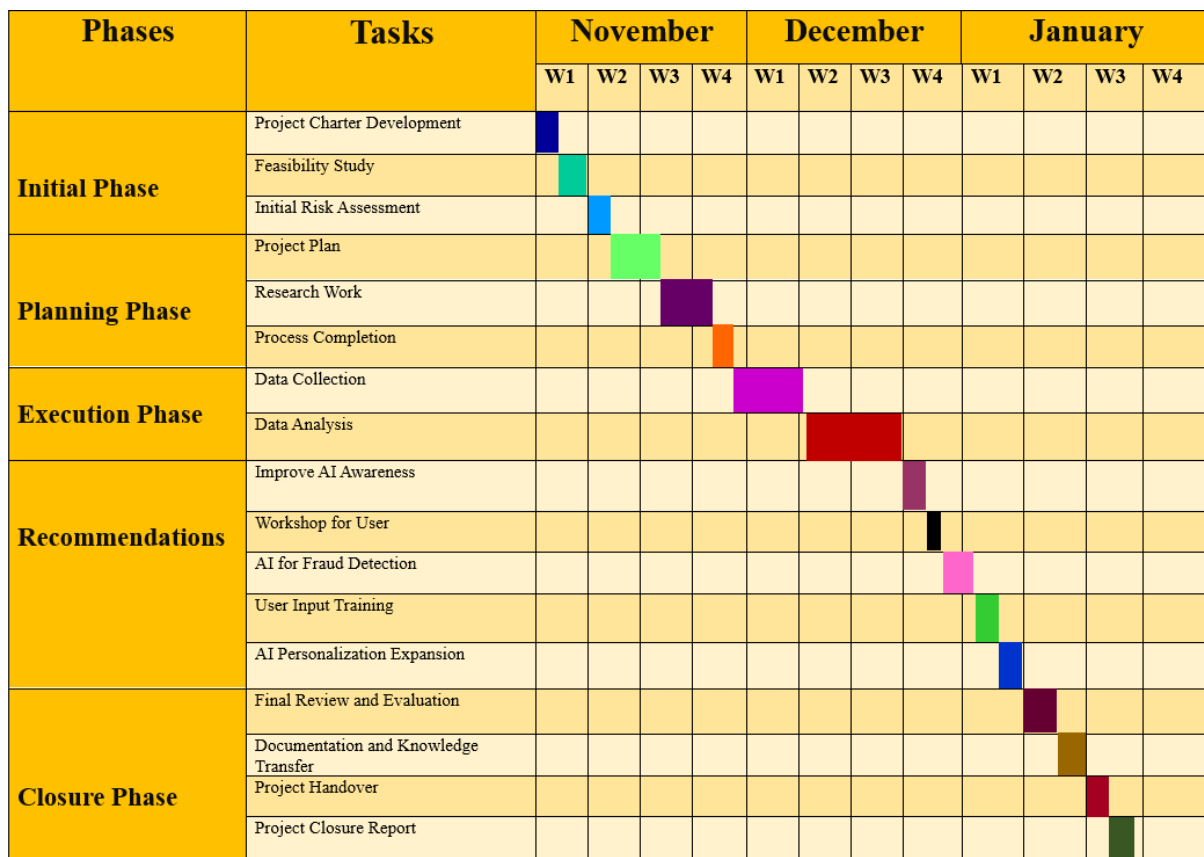


Figure 25 – Revised Gantt Chart

4. Discussion of the Accuracy and Reliability of the different research methods applied

4.1. Introduction to the Accuracy and Reliability

Accuracy

Accuracy refers to the degree to which the research findings accurately reflect the actual phenomena under investigation. In this study, accuracy is ensured by adopting structured research methods, carefully designed data collection tools, and validated analytical techniques. It aims to reduce errors and provide a precise representation of the customer experience with Araliya Bank's AI-powered online transaction platform.

Reliability

Reliability refers to the ability of the results to be repeated over time, under different circumstances. It allows the results obtained to be actual and not dependent on chance. This

study insists on reliability since it uses appropriate methods, a survey instrument with no changes in design, and strictly controlled procedures at every stage.

4.2. Accuracy and Reliability of the different research methods applied

Research methodology used

This mixed-methods study collects data from both qualitative and quantitative methods, aiming for comprehensiveness. The additional of the qualitative methods adds depth and context to the findings, while the quantitative methods ensure that the numbers are precise. This duality ensures accuracy and increases reliability since the diverse dimensions in customer experiences get captured.

Sample Size and Sampling Techniques

A representative sample size of active customers using the AI-powered online transaction platform provided by Araliya Bank is chosen. The stratified random sampling will ensure that the sample represents all demographic and behavioral customer segments, increasing the accuracy of the findings. Sampling techniques are planned to reduce the risk of selection bias and increase reliability.

Data Collection Methods

Primary data is collected through structured questionnaires and customer surveys, while secondary data is obtained from verified reports, including the annual stakeholder reports of Araliya Bank. This will ensure that the data is accurate and reliable since standardized and validated tools are used in data collection.

Preparation of Questionnaire

The questionnaire is designed to be highly structured, covering all aspects of customer experience, such as usability, efficiency, security, and satisfaction. Questions are cleared ambiguity and are neutral in respect to the possibility of biased answers. Their relevance is tested to ensure it is high. Further, its reliability is verified through piloting with a small subnet of respondents.

Data Analysis Techniques

The data is interpreted using statistical and thematic analysis methods. While theme analysis is used to consistently categorize patterns in quantitative data, statistical software is used to assure precision in quantitative data analysis. The quality and dependability of the study findings are improved by these exacting analytical techniques.

5. Evaluation of Project Recommendations with Organizational Needs

To ensure that the project recommendations effectively address the needs of Araliya Bank, each recommendation was evaluated based on its alignment with organizational goal and the rationale for the planning decisions made.

Project Recommendation	Organizational Needs Addressed	Rationale for Planning Decision
Implementation of AI-powered fraud monitoring and detection.	Identifying and stopping fraud in online transactions.	Fraudulent transactions are detected and stopped by real-time AI-powered monitoring, boosting security and confidence.
Personalization of digital banking services using AI	Improved customer satisfaction and retention.	AI-driven recommendation of tailored financial products and services based on customer behaviour improves satisfaction and loyalty.
Predictive analytics integration for consumer insights.	Making decisions based on data to meet client demands.	Better decision-making and more focused products are made possible by AI analytics, which give actionable insights into consumer behaviour.

Transaction workflow optimisation using AI-powered technologies	The effectiveness of digital banking operations	AI speeds up processing times and lowers operating expenses by automating repetitive processes.
Deploy AI-based cross selling tools.	Revenue maximization through personalized recommendations.	Upselling and cross-selling opportunities are identified by AI, so revenue increases while customer interaction improves.

Table 10 – Evaluation of Project Recommendations with Organizational Needs

The goals determined for completing this project during the planning phase are referred to as project planning recommendations. Araliya Bank can successfully increase its operational efficiency through AI technologies by coordinating these recommendations with organizational goals and guaranteeing the precision dependability of the research methodologies used. The recommendations are realistic, actionable and customized to the particular requirements of the company thanks to the through planning and thorough investigation.

6. Assessing Project Planning Recommendations

Project Planning Recommendations	Organizational Needs Addressed	Accuracy and Reliability of Research
Implementation of AI-powered of AI-powered fraud monitoring and detection	Identifying and preventing fraud in online transactions	The study uses past transaction data and user reviews to assess how well AI detects fraudulent transactions
Personalization of digital banking services using AI	Customer satisfaction and customisation of services	The study actual data to show how AI-powered personalisation boosts customer engagement and satisfaction

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Predictive analytics integration for consumer insights.	Compliance and risk management	The study carefully assesses AI's prospective risk prediction abilities, backed by case studies and real-world application.
Transaction workflow optimisation using AI-powered technologies	Cost savings and operational effectiveness	The study offers trustworthy indicators that show how AI automation lowers operating expenses and transaction processing times.
Deploy AI-based cross selling tools	Improved customer service and answering of questions	Through response time analysis and customer satisfaction survey, the study evaluates the efficacy of chatbots and demonstrates their dependability.

Table 11 – Assessing Project Planning Recommendations

Conclusion

I found this assignment to be challenging yet rewarding. I learned more about impact AI has on industries and the importance of ethical and strategic considerations in its implementation while researching the practical applications and limitations of AI. I did project planning, stakeholder role, and risk associated with the adoption of AI that helped me enhance my skills in researching and analyzing. I believe this insight and these skills will carry forward to future academic and professional undertakings in life.

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